

AKADEMIKA

FOL-PHYSICS

PHYSICS OF FIBER OPTIC LAB

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. FIBER OPTIC COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate

Opp. Karishma Society, Off Karve Road,

Kothrud, Pune, Maharashtra – 411038

INDIA

Tel. +91-9004904462, +91-7447438443

E-mail : info@akademika.in

www.akademika.in

SAFETY RULES

Read carefully and follow the instructions mentioned in this manual. This use manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.

After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also ensure that the accessories have no visible damage.

Laser Safety

It is imperative that you read and understand the following safety information before attempting any of the experiments detailed here if you have any enquiries please consult your lecturer or demonstrator

Operational Hazard-Semiconductor Laser Diode

The laser diode sources emit radiation, which is visible to the human eye and may cause **eye damage** if the output beam is viewed directly. When use, never stare at the optical output port of the Laser Diode when its dust cap is removed. Always leave the dust caps for the fiber patch cord in position until ready to make a connection and never stare at the end of a fiber patch cord connected to the Laser Diode output port or point the fiber end at yourself or anyone else in your vicinity before making a connection. Never use external optics (i.e. collimating lenses) to view the Laser Diode emission at the close range.

The output power of the semiconductor Laser Diode is less than 1mW conforms to all applicable standards of DHMS regulations and falls within international safety Class3B.

This Lab must be employed only for the use for which it has been conceived, i.e. as an educational system kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal or careless use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well. For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

SR. NO	CHAPTER	PAGE NO
01.	Technical Specification	05
02.	Functional Block	07
03.	Introduction	08
04.	Experiment No. 1 Mode observation in optical fiber	29
05.	Experiment No.2 Measurement of Numerical Aperture	35
06.	Experiment No.3 The Fiber Attenuation by Fiber Cutback Method	43
07.	Experiment No.4 Coupling efficiency of a Laser to Fiber Coupler	49
08.	Experiment No. 5 Measurement of the Mode Field Diameter of an Optical Fiber	55
09.	Component Tour	66

TECHNICAL SPECIFICATIONS

- He-Ne laser source with 1mW output power and 633.5nm wavelength
- Laser to fiber coupler with Lens adjustment facility to adjust beam into core of fiber, Coupling Efficiency of >70% for SM fibers and > 90% for MM fibers, Wavelength of operation 180 to 2000nm, power handling capacity more than 1 watt
- Laser power meter with separate sensor unit with stand and separate display unit, power measurement up to 40mW. Si detector with 9 mm diameter. Wavelength of operation 400nm - 1100 nm and calibrated to 633nm.
- Optical fiber patch chords and accessories required for experimentation purpose
 - 9 μ m single mode glass fiber patch chord
 - 62.5 μ m multimode glass fiber patch chord
 - 9 μ m single mode glass fiber pigtail
 - FOL-FIBER unit with multimode fiber of 100m, 500m and 1000m length.
 - ST to ST Matting Sleeve –2
 - Optical platform
 - Laser to fiber coupler holder
 - Laser Source holder
 - Fiber Holder with vertical and horizontal positioning facility
 - Rotary stage with angle rotation in steps better than $\frac{1}{2}$ degree.
 - Display screen

- **Deliverables**

Optical Bread Board	: 01
He-Ne Laser Source with Power Supply	: 01
Laser Power Meter	: 01
Gimble Mount for He-Ne Laser	: 01
Gimble Mount Block	: 01
X-Y-Z Stage	: 01
Coupler Holder	: 01
Laser to fiber Coupler	: 01
Rotary Stage	: 01
Rotary Stage Block	: 01
Rotary Fiber Holder with Receptacle	: 01
Fiber Coupler with Groove Fiber holder	: 01
Fiber Coupler with Groove Fiber holder Block	: 01
Photo detector cap With Receptacle	: 01
Photo detector cap With Knife-edge Slit	: 01
Photo detector Holder adjustable stand	: 01
Fiber Holder Stand	: 01

- **Fiber Cable**

(A) 9/125 μm 1Meter Optical Patch Cord	: 02
(B) 62.5/125 μm Optical Patch Cord	: 02
(C) 62.5/125 μm 5m Optical Pigtail	: 01
(D) 100/140 μm 1m multimode Optical Pigtail	: 01
FOL-Fiber containing 100 meter, 500 meter and 1000 meter 62.5/125 μm multimode fiber cable	: 01

- **Allen Key Set**

(A) M2	: 01
(B) M3	: 01
(C) M5	: 01
Display Screen	: 01

- **Allen Bolt**

(A) M6 X 10 mm	: 06
(B) M6 X 15mm	: 06
(C) M4 X10 mm	: 04
Experimental Manual	: 01

FUNCTIONAL BLOCK



INTRODUCTION

Fiber optics technology is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All big telephone companies like AT & T, sprints have saturated the airways & print media with advertisements heralding this & new technology. Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

And their enthusiasm is valid. Today medical fiber optics systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologists ever come close to the fantastic growth rate that market analysts love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will pervade our homes workspaces & recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronics newspapers become feasible. Two-way fiber optics will allow you to “attend college” in the comfort of your home.

Anyone who takes an interest in fiber optics & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few experts exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons existing with the prospect of making new discoveries finding new application for the technology.

These modules are an introduction to fiber optics communication technology for instructors, students & hobbyist. Welcome to fascinating & expanding world of fiber optics. We hope that you find these modules are challenging, stimulating & yes even fun.

THEORY OF FIBER OPTICS

What is fiber-optic communication?

A fiber-optic communication system is a communication system in which the transmitter and receiver are linked by fiber-optic cable. At the transmitter, light is shone into the cable. This travels down the fiber-optic cable, and can be detected at the receiver.

If we change the brightness of the light beam at the transmitter, we can detect this change in brightness at the receiver-this allows us to send information over the fiber-optic link.

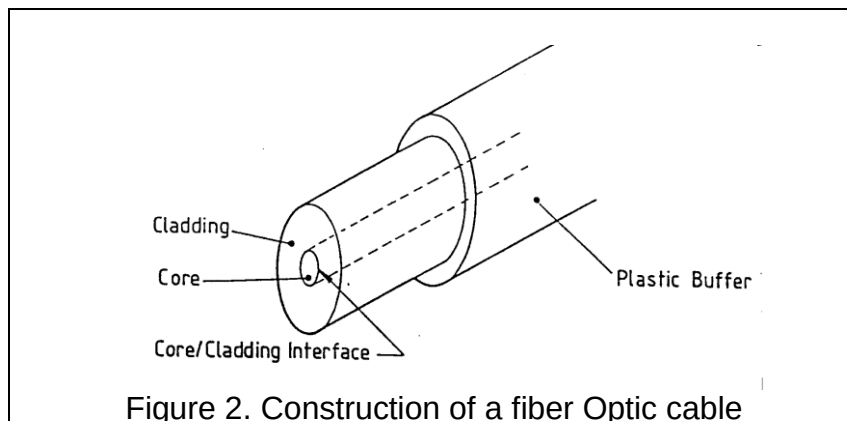
Why use light?

Fiber-optic cables have a number of advantages over metal cables when it comes to communication systems. They:

- Have smaller diameters
 - Are lighter in weight and more flexible
 - Allow much greater distance to be covered without any amplifiers / electronics
 - Have extremely large bandwidths
 - Provide electrical isolation between transmitter and receiver
 - Naturally insulate between any electromagnetic wave energy
- (i.e. it is extremely difficult to eavesdrop)
- Are immune to interference due to other people's electrical equipment.

What is an optical fiber?

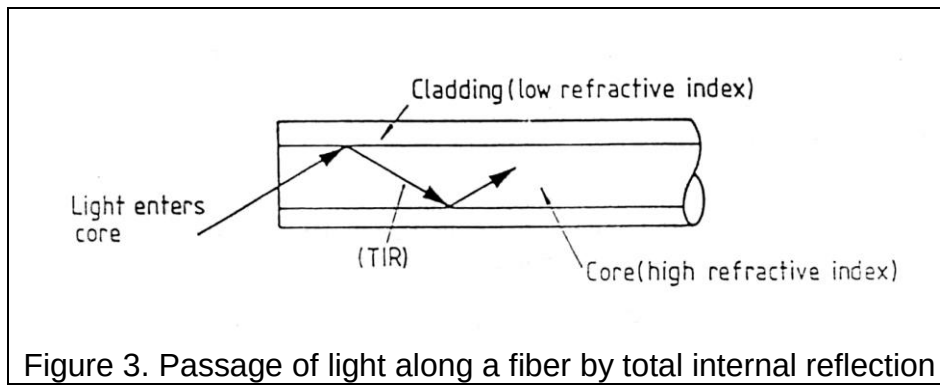
An optical fiber is basically a light wave-guide, which transmits energy at optical frequencies. It consists of a central transparent core down which the light travels, surrounded by a layer of transparent cladding, which creates an optical interface at the core/cladding boundary as shown in Figure 2:



The core and cladding may be made from either **glass** or **plastic**, and non-transparent **plastic buffer** to provide environmental protection surrounds the cladding.

How does light travel down an optical fiber?

Light is transported through the fiber-optic core by continuously being reflected off the core/cladding interface. This is called **total internal reflection (TIR)**, and is shown in Figure3 below:

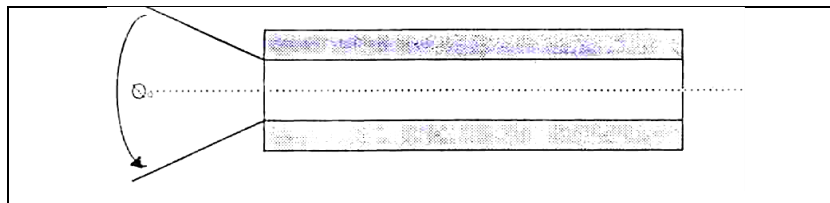


The reflections are made possible by having a high refractive index for the core material and a lower refractive index for the cladding material.

To create TIR the core must have a higher refractive index than the cladding, and as we shall see later, the light must not strike the core/cladding interface at too steep an angle.

Acceptance angle and numerical aperture

In order for total internal reflection to take place inside an optic fiber, light rays entering the end of the fiber must fall inside an imaginary 'cone' of angle Φ° , as shown below



Although light rays outside this cone may still enter the fiber, they will hit the core-cladding interface at too steep an angle for TIR to take place, and their energy will be lost into the cladding.

At the far end of the fiber, light will emerge in a similar cone, also of angle Φ .

The angle Φ is an important parameter of any fiber-optic cable, and is called the **acceptance angle** of the fiber.

To avoid conversion over units (degrees, radians, etc.), the term numerical aperture is often used in data sheets for optic fibers. The numerical aperture (NA) of a fiber is defined as the sine of half the acceptance angle, i.e.

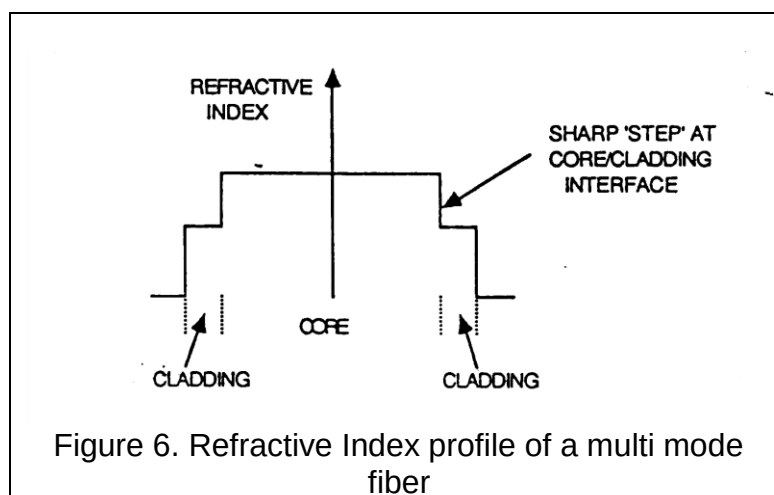
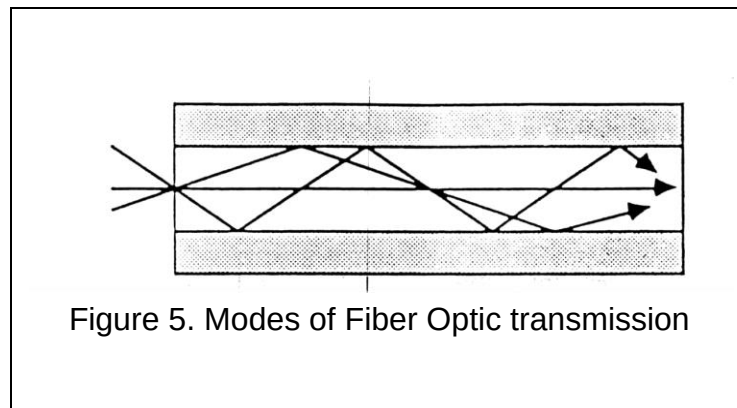
$$NA = \sin (\Phi/2)$$

Multimode Fibers

Optical fibers can be divided into two main classes - **multimode** and **single mode**

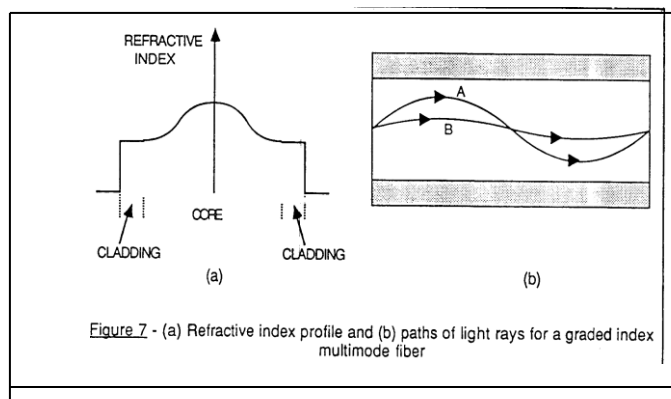
In multimode fibers, the diameter of the cable's core is large enough to allow different light rays to take different paths as they travel along the cable. Some will be continually reflected off the core-cladding interface, while others may travel near the center of the core, hitting the core-cladding interface only occasionally.

These different paths are shown in Figure 5:



Note from Figure 6 that the refractive index throughout the core is constant: - this means that all light rays travel at the same speed though the core. Since the distance traveled by each light ray depends on how often it is reflected off the core cladding surface (see figure 5), this means that different light rays will arrive at the end of the cable at different times, depending on the paths the rays take through the fiber. This results in a spreading or **modal dispersion** of the light reaching the end of the fiber, which limits the bandwidth available for communication.

To minimize the problem of modal dispersion in high- transmission –rate communications, another type of multimode fiber, known as a **graded index** fiber, was developed. With this type of fiber, the refractive index of the core is not constant, but decreases gradually as the distance from the core's center increases. Consequently, there is no sharp change in refractive index at the core/cladding interface - the refractive index of the cladding is approached gradually, as shown in figure 7 (a):



The speed of each light ray depends on the refractive index of the material through which it passes- the lower the refractive index, the faster the light ray will travel. Rays traveling close to the core's center have the least distance to travel, but the refractive index in this region is high, so their speed will be relatively low.

Conversely, those rays traveling near to the cladding have the longest distance to travel, but since the refractive index in this region is low, their speed will be relatively high.

For example, in figure 7 (b) above, ray A has the longest path, but it spends more time in regions of lower refractive index than ray B – hence the average speed of A is higher. By properly designing the fiber, the transit times of these and all other rays can be made almost identical.

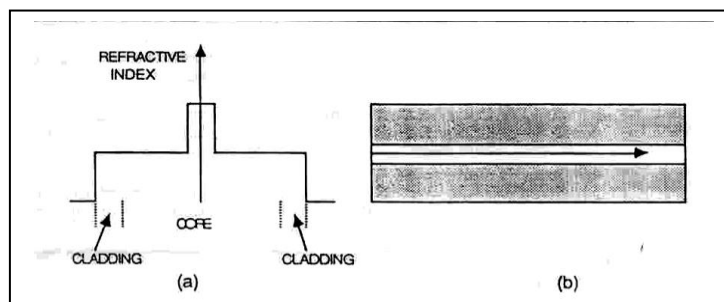
Thus, with careful design, all rays will arrive at the end of the fiber at more or less the same time. In this way, graded index multimode fiber minimizes the problem of modal dispersion.

Single-Mode fibers

Single mode (or 'monomode') fibers allow communication at faster data rates, and over longer distances, than is possible even with graded index multimode fiber.

Although single-mode fibers are similar to multimode step-index fibers in their construction, they have much smaller core diameters. In fact, the diameter of the core is so small that there is only one mode in which light rays can travel down a single mode fiber – that is, straight down the core, only just grazing the core-cladding interface. It is this mode of transmission, which gives the single-mode fiber its name.

The refractive index cross-section of a single-mode fiber and the path taken by all light rays in this type of fiber, are shown in figure 8 parts (a) and (b) respectively.



Since single-mode fibers only support one transmission mode, they do not suffer from the modal dispersion problems, which limit the bandwidth of multimode fibers. Consequently, they are mainly used in high-transmission-rate, long distance applications.

Dispersion in fiber-optic cables

Dispersion is caused by light rays from the same light source taking different lengths of time to travel along a fiber-optic cable. If a short pulse of light is sent down a long optic fiber, dispersion can result in a 'spreading' of the pulse received at the far end of the cable. This phenomenon is known as **pulse spreading**, and is shown in figure 9:

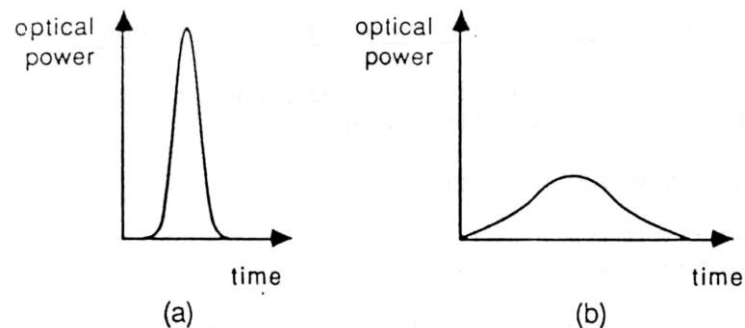


Figure 9. Pulse spreading: a). Transmitted B) Received pulses

Since dispersion prevents the successful transmission of fast pulses of light, its overall effect is to limit the bandwidth of a fiber-Optic cable.

There are two main types of dispersion, which occur in optic fibers - modal dispersion, and material dispersion.

It occurs when individual light rays take different paths during their passage through the fiber's core. This leads to some light rays traveling further than others, so that some rays may take longer than others to reach the far end of the cable.

Material dispersion is related to the fact that different wavelengths of light travel at different speeds in the same medium. This means that different light wavelengths, launched simultaneously into a fiber from a light source, will not arrive at the receiver at the same instant in time, because some will travel more slowly than others. Then result is dispersion due to the differences in travel time.

Using a light source, which only generates light over a very narrow range of wavelengths, can reduce the problem of material dispersion. Such a narrow emission spectrum is only obtainable from a laser.

Dispersion is only a problem in long distance, fast-transmission-rate communications. In such applications, single mode or graded-index multimode fibers are used in conjunction with lasers, to keep both modal and material dispersion to a minimum.

Transmission Losses in fiber-optic cables

Transmission loss is a significant feature of any optical fiber, since it determines how much light reaches the far end of a given length of cable.

There are four main causes of transmission loss, which contribute to the overall attenuation of the fiber, in dB/km. They are:

- (a) Material absorption
- (b) Material scattering
- (c) Scattering due to irregularities at the core/cladding interfaces
- (d) Radiation due to cable curvature

Material absorption results from molecular impurities within the core of the fiber, which absorb light at certain wavelengths. This gives rise to local attenuation peaks within the cable, if transmission loss is plotted against the wavelength of light used:

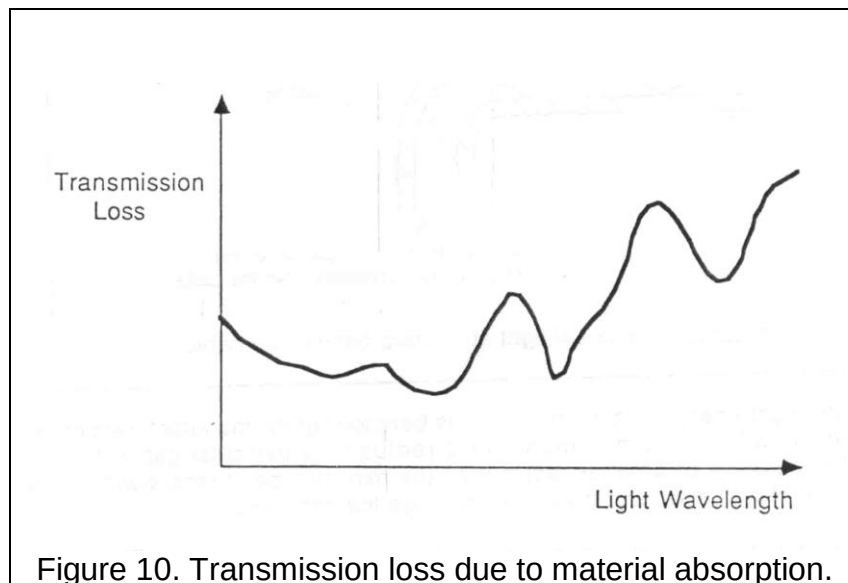


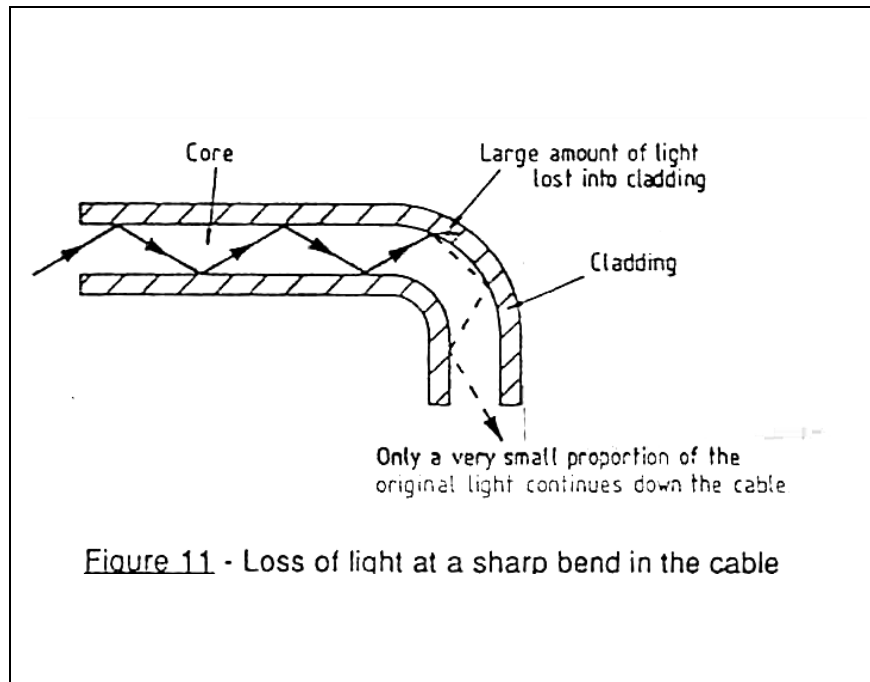
Figure 10. Transmission loss due to material absorption.

To minimize losses due to material absorption, an emitter device is chosen which has a peak output wavelength corresponding to a spectral region where cable attenuation is a minimum. This occurs in the 'troughs' of the graph in Figure 10.

Material scattering is caused by particle impurities and by fluctuations in temperature and core composition, which interrupt the reflection paths of the light rays traveling down the cable. The result is a loss of optical power due 'to scattering' of the light rays.

Further scattering can arise due to irregularities at the **core/cladding interface**, resulting in light energy being lost into the cladding.

Radiation due to cable curvature also results in cable losses. If the curvature at any point along the cable's length is too great, some light rays will strike the core/cladding interface at too steep an angle. Much of the energy contained in these light rays will be lost into the cladding, and only a small proportion will be reflected back into the core, as illustrated in Figure11:



Because light energy is lost if the cable is bent too tightly, manufacturers of fiber-optic cable will quote a minimum bend radius for a particular cable. Putting a bend in the cable of smaller radius than the minimum bend radius will cause appreciable light loss, and may even damage the cable itself

Later in this manual, you will investigate how transmission loss increases with distance, and you will calculate the transmission loss, in dB per meter, of the cable supplied.

Coupling Losses

In addition to losses inherent in the fiber, optical power is also lost in coupling light into and out of, the fiber cable. There are three main situations where coupling losses occur:

- In launching light into a fiber from the emitter.
- In coupling light from a fiber onto the photo detector.
- In transferring light from one length of fiber to another.

The main causes of coupling losses with this type of cable are:

- (a) Gap separation between fiber ends
- (b) Connector misalignment
- (c) Poor-quality fiber end terminations

The ends of all fiber-optic cables supplied have been pre-polished at the factory, to ensure that losses due to (c) above are minimized.

GEOMETRICAL OPTICS AND FIBER OPTICS

Light As An Electro Magnetic Field

The light by which we see the world around us is a part of the range, or spectrum, of electromagnetic waves that extends from radio frequencies to high power gamma radiation. These waves, a combination of electric and magnetic fields, which can propagate through a vacuum, have as their most distinguishing features their wavelength and frequency of oscillation. The range for wavelengths for visible light is from about 400 nanometers (nm) to about 700 nm. In most of the work done in the field of fiber optics, the most useful sources of electromagnetic radiation emit just outside the visible in the near infrared with wavelengths in the vicinity of 800 to 1500 nm.

It can be difficult to follow what happens in an optical fiber system if the progress of light through the system is depicted in terms of the wave motion of the light. For the simplest cases it is easier to think of light traveling as a series of rays propagating through space. In a vacuum, light travels at approximately 3×10^8 meters per second. In material media, such as air or water or glass, the speed is reduced. For air, the reduction is very small; for water, the reduction is about 25%; in glass, the reduction can vary from 30% to 50 %.

Light In Materials

In most cases, the results of the interaction of an electromagnetic wave with material medium can be expressed in terms of a single number, the index of refraction of the medium. The **refractive index** is the ratio of the speed of light in a vacuum, c , to the speed of light in the medium, v ,

$$n = c / v \quad (0-1)$$

Since the speed of light in a medium is always less than it is in a vacuum, the refractive index is always greater than one. In air, the value is very close to one; in water, it is about 1.33, in glasses, it varies from about 1.44 to about 1.9. The refractive index varies with the wavelength of the light. This is called **wavelength dispersion**. Also, not only can the medium slow down the light, but it can also absorb some of the light as it passes through.

In a homogeneous medium, that is, one in which the refractive index is constant in space, light travels in a straight line. Only when the light meets a variation or a discontinuity in the refractive index will the light rays be bent from their initial direction.

In the case of variation in the refractive index within a material, the behavior of the light is governed by the way in which the index changes in space. For example, the air just above a road heated by the sun will be less dense than the air further from the road. Since the refractive index increases with density, the refractive index of the air increases with height. This is called **refractive index gradient**. The light passing through such a medium of varying refractive index will undergo bending.

This gradual bending of light by a refractive index gradient is used in fiber optics to increase the information carrying capacity of fibers and to provide a very compact lens for fiber optic systems.

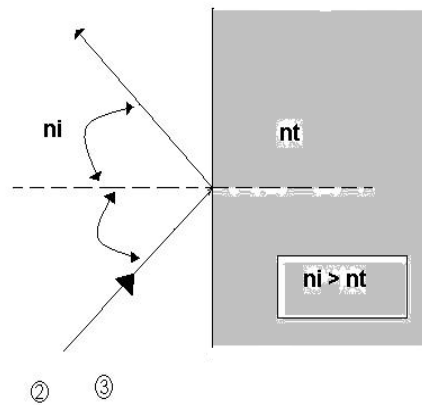


Figure 0.6. geometry of reflection and refraction

If the change in refractive index is not gradual, as in the case of the refractive index gradient, but is, instead, an abrupt change like that between glass and air, the direction of light is governed by the Laws of Geometrical Optics. If the angle of incidence Θ_i , of a ray is angle between an incident ray and a line perpendicular to the interface at the point where the light ray strikes the interface (**Fig 0.6**), then :

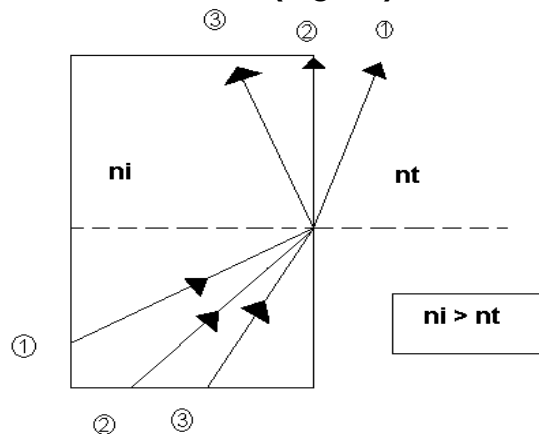


Figure 0.7. Geometry Of Total Internal Reflection Ray#1 Is At Less Than The Critical Angle. Ray #3 Is Totally Internally Reflected

1. The angle of reflection, Θ_r , also measured with respect to the same perpendicular, is equal to the angle of incidence:

$$\Theta_r = \Theta_i \quad (0-2)$$

2. The angle of the transmitted light is given by the relation:

$$n_t \sin(\Theta_t) = n_i \sin(\Theta_i) \quad (0-3)$$

The first of these relations is known as the **Law of Reflection**; the second as the **Law of Refraction** or **Snell's Law**.

It is useful to refer to a material whose refractive index is greater than another as being optically denser and one whose refractive index is less as being rarer. Thus, light traveling into an optically denser medium would be bent toward the normal, while light entering an optically rarer medium would be bent away from the normal. In **Fig 0.7**, a series of rays in a dense medium are incident on an interface at different angles of incidence. Ray #1 is refracted at the interface to a rarer medium according to Snell's Law. Ray #2 is incident at an angle such that the refracted angle is 90 degrees. Ray #3

is incident at an even larger angle. If the angle of incidence of Ray #3 is inserted in Snell's Law, the sine of the angle of transmission will be found to be greater than one! This cannot happen. Instead, all of the light is reflected back into incident medium. There is no light transmitted into the second medium. The light is said to be **totally internally reflected**. For angles of incidence greater than a critical angle, total internal reflection will occur. This **critical angle** occurs at the angle of incidence at which the transmitted ray is refracted along the surface of the interface (the case illustrated by Ray #2). Setting the angle of transmission equal to 90 degrees, the critical angle, θ_{crit} is found from Eq. 0-3 to be:

$$\sin(\theta_{crit}) = n_t / n_i \quad (0-4)$$

In the ray picture, the concept of total internal reflection makes the interface look like a perfect mirror. If the process is examined in terms of wave propagation, theory predicts and experiments confirm that a weak electromagnetic field exists in rarer medium, but it decays rapidly with distance from the interface and no light energy is transmitted to the rarer medium. This field is called an **evanescent field**. However, if another optically dense material were located very close to the material in which the total internal reflection were occurring (within a wavelength or so) some of the light energy could be coupled out of the first medium across the small gap and into the second dense medium. This process is called **frustrated total internal reflection** since the usual reflection is frustrated by the location of the material next to the interface. Frustrated total internal reflection is responsible for the operation of a component in fiber optic systems called a bidirectional coupler.

Light in Optical Fibers

If one were to use a fiber consisting of only a single strand of glass or plastic, light could be lost at any point where the fiber touched a surface for support. Thus, the amount of light that could be transmitted would be dependent on the methods used for holding the fiber. The output of the fiber would also be affected by any movement of the fiber during its use. To eliminate these problems, the central light carrying portion of the fiber, called the **core**, is surrounded by a cylindrical region, called the **cladding** (Fig 0.8). The cladding is then covered with a protective plastic jacket. Since the refractive index between the core and the cladding is less than in the case of a core in air, the critical angle is much bigger for the clad fiber. The index of the cladding, n_{cl} , is still less than the index of the core, n_{core} , because total internal reflection will occur only when $n_{core} > n_{cl}$. Looking at a cross-section of the fiber in Fig 0.8, one sees that the cone of the rays that will be accepted by the fiber is determined by the difference between the refractive indices of the core and cladding. The **fractional refractive index difference** is given by

$$\Delta = (n_{core} - n_{cl}) / n_{core} \quad (0-5)$$

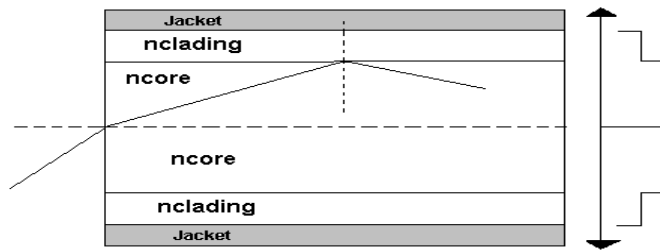


Figure 0.8 Step index-fibers. The refractive index profile is shown at the right. The geometry for derivation of the numerical aperture is given.

Because the refractive index of the core is a constant and the index changes abruptly at the core-cladding interface, the type of fiber in **Fig 0.8** is called a **step-index fiber**.

The definition of the critical angle can be used to find the size of the cone of light that will be accepted by an optical fiber with a fractional index difference. In **Fig 0.8** a ray is drawn that is incident on the core-cladding interface at the critical angle is θ_c , then Snell's Law.

$$\begin{aligned} n_i \sin \theta_c &= n_{\text{core}} \sin \theta_t = n_{\text{core}} \sin(90^\circ - \theta_{\text{crit}}) \\ &= n_{\text{core}} \cos(\theta_{\text{crit}}) \\ &= n_{\text{core}} \sqrt{1 - \sin^2(\theta_{\text{crit}})} \end{aligned}$$

From Eq.0-4, $\sin(\theta_{\text{crit}}) = n_{\text{cl}} / n_{\text{core}}$, so

$$n_i \sin \theta_c = \sqrt{n_{\text{core}}^2 - n_{\text{cl}}^2} \quad (0-6)$$

The **numerical aperture**, NA, is a measure of how much light can be collected by an optical system, whether it is an optical fiber or a microscope objective lens or a photographic lens. It is the product of the refractive index of the incident medium and the sine of the maximum ray angle.

$$NA = n_i \sin(\theta_{\text{max}}) \quad (0-7)$$

In most cases, the light is incident from air and $n_i = 1$. In this case the numerical aperture of a step index fiber is, from Eq. 0-6 and 0-7

$$NA = \sqrt{n_{\text{core}}^2 - n_{\text{cl}}^2} \quad (0-8)$$

When $\Delta \ll 1$, Eq.0-8 can be approximated by

$$\begin{aligned} NA &= \sqrt{(n_{\text{core}} + n_{\text{cl}})(n_{\text{core}} - n_{\text{cl}})} \\ &= \sqrt{(2n_{\text{core}})(n_{\text{core}} \Delta)} = n_{\text{core}} \sqrt{2 \Delta} \end{aligned} \quad (0-9)$$

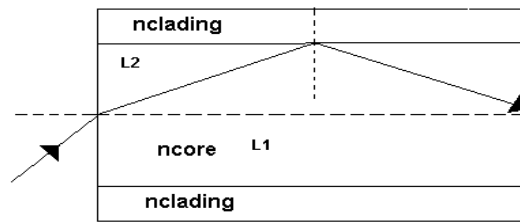


Figure 0.9. The geometry for derivation of the differential delay of a step-index fiber

The condition in which $\Delta \ll 1$ is referred to as the **weakly-guiding approximation**. The NA of fiber will be measured in one of our experiments.

In **Fig. 0.9**, two rays are shown. One the **axial ray**, travels along the axis of the fiber; the other, the **marginal ray**, travels along a path near the critical angle for the core-cladding interface and is the highest-angle ray which will be propagated by the fiber. At the point where the marginal ray hits the interface the ray has traveled a distance L_2 , while the axial ray has traveled a distance L_1 . From the geometry, it can be seen that

$$\sin \Theta = n_{cl} / n_{core} = L_1 / L_2 \quad (0-10)$$

The length L_2 is a factor n_{cl} / n_{core} larger than L_1 , in the case shown in the figure. For any length of fiber L , the additional distance traveled by a marginal ray is

$$\delta L = (n_{core} - n_{cl})L / n_{cl} \quad (0-11)$$

Eq. 0-11 can be simplified to $\delta L = L\Delta$. The additional time it takes light to travel along this marginal ray is

$$\delta t = \delta L / v = L\Delta n_{core} / c \quad (0-12)$$

Therefore, a pulse with a length t representing one bit of information will be lengthened to $t + \delta t$. This **differential time** between the axial and marginal rays will cause a pulse to smear and thereby limit the number of pulses per second that could be sent through a fiber and distinguished at the far end. In such a case, the system may be limited not by how fast the source can be turned on and off or by the speed of the detector, but by the differential time delay of the fiber. This smearing of pulses can be remedied through the use of the graded-index or single-mode fibers.

Earlier, it was noted that rays can be deflected by variations in the refractive index of a medium as well as by encountering an abrupt interface between two indices. There are a number of methods of creating controlled index gradients. Some involves introducing impurities into thin layers of glass as they are laid down on a substrate. This is not a continuous process since the refractive index within each layer is nearly constant. The resulting variation of refractive index in a fiber resembles that of a series of concentric tree rings rather than a smooth change in the index. Other techniques involve the removal of material from the base glass by some type of chemical method. Fibers whose cores have such an index gradient are called **graded-index fibers**. In our discussion, we will not make a distinction between these processes, but instead assume that the graded-index profiles are smooth and exactly conform to theory. In real graded-index

optical fibers this may not be correct and such departures from the ideal gradient will affect their performance.

Once the refractive index gradient can be controlled through manufacturing processes, it is up to the designer of optical fibers to determine the most useful **refractive index profiles**, $n(r)$, and the variation of index with radial distance in the core. This usually fits a power law profile given by

$$n^2(r) = n^2 [1 - 2(r/a)^\alpha], \quad (0-13)$$

Where n_0 is the index of refraction at the center of the core, and Δ is the fractional index difference defined earlier in Eq. 0-5, but with n_0 now substituting for n_{core} .

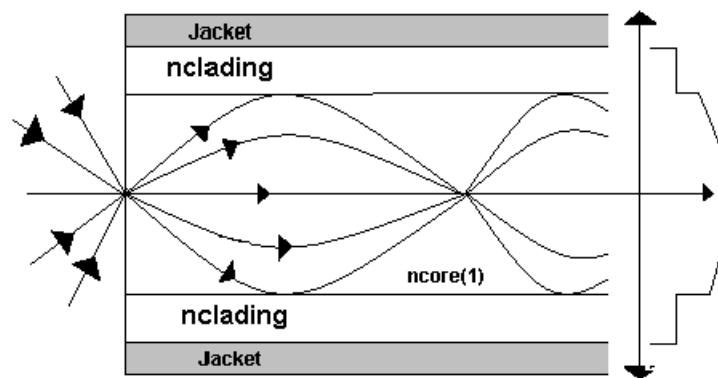


Figure 0.10. Graded-index fiber The refractive index profile is shown at the right angle. Diverging rays are refocused at a point further down the fiber.

The parameter α , is the exponent of the power law and determines the shape of the graded-index profile. When $\alpha = \infty$ the profile is that of a step-index fiber. For $\alpha = 2$, the profile is parabolic (**Fig 0.10**). This is the profile found in most telecommunications graded-index fibers, because this profile eliminates the differential time delay between axial and marginal rays. The numerical aperture of a graded-index fiber is the same as that of a step-index fiber only for rays entering on the fiber axis. For rays entering at other points on the core, the local numerical aperture is less because the local index, $n(r)$, must be used in Eq. 0-8. In the case of the parabolic graded-index fiber, the total amount of light which can be collected is one half of that which can be controlled by a step-index fiber with the same Δ .

Without plunging in the mathematics needed to prove that graded-index fibers with a parabolic profile remove differential delay, it is possible to get a qualitative feel of why the smearing of light pulses in such a fiber would be reduced. Instead of the rays bouncing off of the core-cladding interface as in step-index fibers, rays follow a gently curving path in graded-index fibers. In those with a parabolic profile, this path is sinusoidal. That is, the path can be described as a sine function in space. It would seem that light paths which have large radial amplitude are still longer than the path down the axis of the fiber. But because of the refractive index gradient, the velocity of the light at the center of the fiber is smaller than the velocity of the light nears the edge of the core. Although the light

that travels near the edge of the fiber has to go farther, it travels faster and arrives at the end of the fiber at the same time as the light traveling down the center of the fiber. If the length of the fiber is L and the speed of light down the center of the fiber is $v = c / n_0$, then the time for a pulse to travel to the end of the fiber is $t = L / v = n_0 * L / c$. For light traveling sinusoidal path, the length traveled will be L'' and the time to travel to the end of the fiber is $t = n(r,z)L''/c$. The product of the geometrical path and the refractive index is called the optical path length. If the optical path length, $n(r)L$, is the same for all paths, there will be no differential delay in the time the rays take to travel through a fiber. For all optical path lengths to be equal, the profile must be parabolic($\alpha = 2$).

0.2.4 Graded-Index Lenses

One thing to note in **Fig. 0.10** is that a fan of rays injected at a point in a graded-index fiber spreads out and then recrosses the axis at a common point just as rays from a small object are reimaged by a lens. The distance it takes for a ray to traverse one full sine path is called the pitch of the fiber. The length of the pitch is determined by Δ , the fractional index difference.

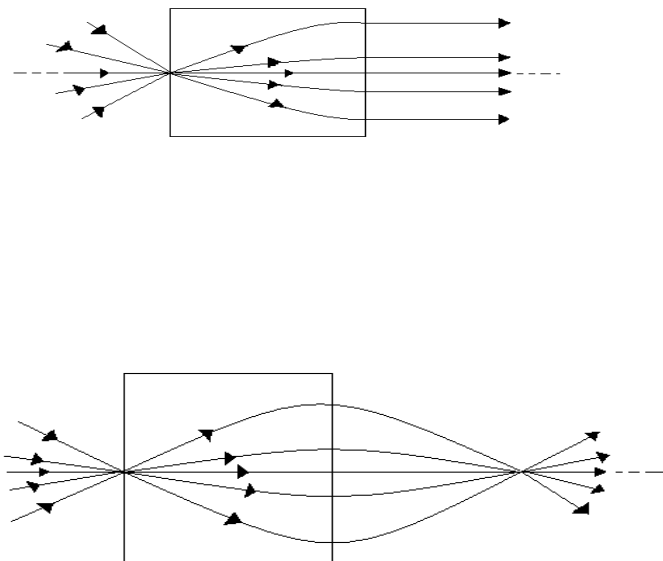


Figure 0.11. Graded-index (GRIN) lens. (a) 0.25 pitch lens. (b) 0.29 pitch lens

If a parabolic graded-index fiber is cut to a length of one quarter of the pitch of the fiber, it can serve as an extremely compact lens (sometimes called a **GRIN lens**, the GRAded-INdex) for fiber applications (**Fig. 0.11**). By positioning the output of a fiber at the face of the short fiber length, light from the lens will be collimated, just as diverging light at the focal point of a lens is collimated. Because its properties are set by its length, this graded-index lens is referred to as a **quarter-pitch or 0.25 pitch lens**.

In some cases, it is not collimation of light which is required, but focusing of the fiber output on to a small detector or focusing of the output of a source on to the core of a fiber. The easiest way of accomplishing this is to increase the length of the GRIN lens slightly to 0.29 of a pitch (**Fig. 0.11**). This enables the fiber optic systems designer to move the source back from the lens and have the transmitted light refocused at some

point beyond the lens. This is particularly useful for coupling sources to fibers and fibers to detectors.

0.3 Wave Optics And Modes In Optical Fibers

Although the ray picture of light propagation through a fiber is easy to depict, it does not reveal some of the interesting properties of light in optical fibers, particularly in those fibers where the core size is of the order of the wavelength of light.

0.3.1 Wave Fields In A Fiber

The laws governing the propagation of light in optical fibers are Maxwell's equations, the same law that describe the propagation of light in a vacuum or any medium. When information about the material constants, such as the refractive indices, and the boundary conditions for the cylindrical geometry of core and cladding is incorporated into the equations, they maybe combined to produce a wave equation that can be solved for those electromagnetic field distributions that will propagate through the fiber. These allowed distributions of the electromagnetic field across the fiber are referred to as the **modes** of the fiber. They are similar to the modes found in microwave cavities and laser cavities. When the number of allowed modes becomes large, as is the case with large-diameter-core fibers, the ray picture we have used gives an adequate description of light propagation in fibers.

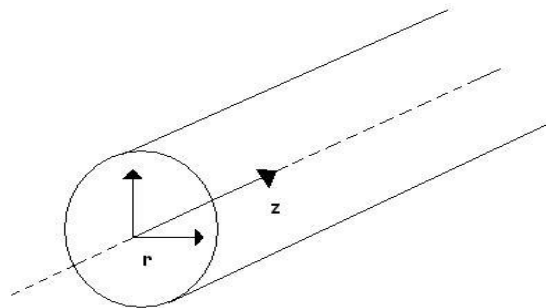


Figure 0.12. Coordinate system for modes in an optical fiber

The description of the modes that propagates in a fiber is found by solving the wave equation in cylindrical coordinates for the electric field of the light in the fiber. The cylindrical coordinate system for a fiber is illustrated in **Fig. 0.12**. The solutions, which are found to be harmonic (consisting of sine and cosine functions) in space and time, are the form

$$E(r, \Phi, z) = f(r) \cos(\omega t - \beta z + \gamma) \cos(q\Phi), \quad (0-14)$$

Where ω is the frequency of light in radian/sec ($\omega = 2\pi\nu$, where ν is the linear frequency in Hertz, cycles/sec), β is the **propagation constant**, expressed in radian per unit distance, γ is a phase constant to provide the correct amplitude at time $t = 0$ and position $z = 0$, and q is an integer. The parameter, β , is important for specifying how light propagates in a fiber. In the ray optics description, β is the projection of the propagation vector on the z axis, where the magnitude of the propagation vector is $k = 2\pi/\lambda_0$, λ_0 being the wavelength of light in vacuum. It is important to make the distinction between

the magnitude of the propagation vector, k , and the propagation constant, β , which is the z – component of the propagation vector, in order to avoid later confusions.

Solutions for β , $f(r)$, and $q(r)$ obtained by substituting **Eq. 0 - 14** into the wave equation. The solutions will depend on the particular fiber geometry and index profile, including both the core and the cladding, under consideration. The step-index profile is one of the few refractive index profiles for which exact solutions may be obtained. For this case the solutions for $f(r)$ are Bessel functions

An important quantity in determining which modes of an electromagnetic field will be supported by a fiber is a parameter called the **characteristic wave guide parameter** or the **normalized wave number**, or, simply, the **V – number** of the fiber. It is written as

$$V = k_f \cdot a \cdot \text{NA}, \quad (0.15)$$

Where k_f is the free space wave number, $2\pi/\lambda_0$, 'a' is the radius of the core, and NA is the numerical aperture of the fiber.

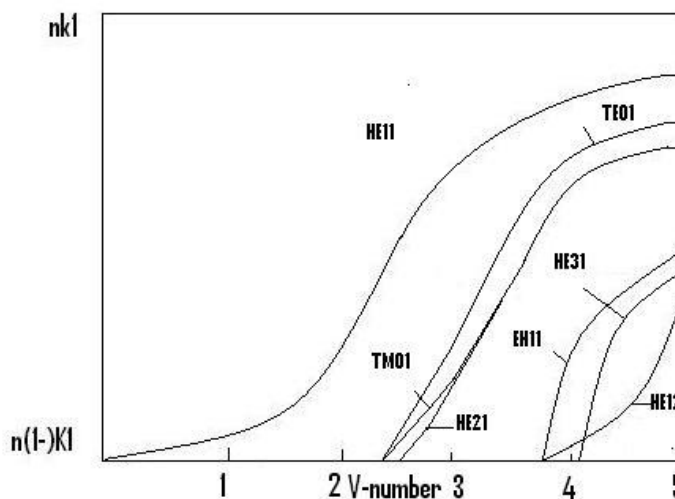


Figure 0.13. Low order modes of an optical fiber. Plot of the propagation constant in a fiber (β) as a function of V-number of a fiber. Each V-number represents a different fiber configuration or a different wavelength of a light in a given fiber configuration.

When the propagation constants (β 's) of the fiber modes are plotted as a function of the V - number (remember, each V – number represents a particular wave number-core diameter-NA product), it is easy to determine the number of modes that can propagate in a particular fiber. In **Fig. 0.13**, such a plot is given for some of the lowest order modes. The number of propagating modes is determined by the number of curves that cross a vertical line drawn at the V – number of the fiber. Note that the fibers with $V < 2.405$, only a single mode will propagate in the fiber. This is the single-mode region. The wavelength at which $V = 2.405$ is called the **cut-off wavelength**, denoted by λ_c , because, for a particular product of core diameter and NA, as the wavelength of radiation is increase, this is the wavelength at which all higher-order modes are cut off and only a single mode will propagate in the fiber. A fiber which propagates only the HE 11 mode is said to be a **single-mode fiber**. For example, the fiber supplied with the kit fiber has a core diameter of $4\mu\text{m}$ and an NA of 0.11. According to Eq. 0-14, this fiber has a V- number of 2.19 for 633nm light, putting it well inside the single-mode region. An experiment with this fiber will be done with the setup provided.

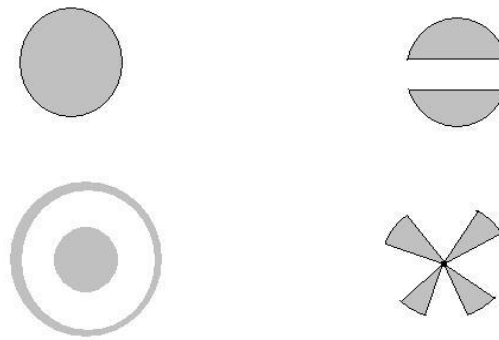


Figure 0.14. Irradiance pattern for some low order linearly polarized modes.

In the weakly-guiding approximation ($\Delta \ll 1$), the exact solutions of waveguide theory, HE_{mn} , can be replaced by a set of modes which are linearly polarized, called the LP modes. (the details of polarization of waves in fibers will be discussed in Sec. 0.3.2.) These LP modes are combinations of the modes found from the exact theory of the waveguide. These linearly-polarized modes may be characterized by two subscripts, m and n . The first subscript, m , gives the number of azimuthal, or angular, nodes (zeroes) that occur in the electric field distribution of the mode; the second subscript, n , gives the number of radial nodes that occur. They can be identified by pattern in the output of the fiber as it illuminates a screen. The patterns are symmetric about the center of the beam and show bright regions separated by dark regions (the nodes that determine the order numbers m and n). Some of these are shown in **Fig. 0.14**. It is assumed that the zero field at the outer edge of the field distribution is counted as a node, so $n \geq 1$. For the azimuthal nodes, $m \geq 0$. The lowest order HE_{11} mode consists of two LP₀₁ modes with polarizations at right angles to one another. **Fig. 0.15** shows the propagation constants of these modes as a function of V - number. (compare this figure with the exact solutions in **Fig. 0.13**.)

When the V - number is greater than 2.405 (the value at which the first zero of the zero-order Bessel function occurs), the next linearly-polarized mode, LP₁₁, can be supported by the fiber, so that both the LP₀₁ and LP₁₁ modes will propagate. For a fiber with a V - number of 3.832 (corresponding to the first zero of the first-order Bessel function), two more linearly-polarized modes can propagate: the LP₂₁ and the LP₀₂ modes. By changing the position and angle of the input beam incident on a low- V number multimode fiber, individual linearly-polarized modes can be launched in the fiber and observed at the output. The propagation of individual modes in such a fiber will be observed in our experiment. This will help overcome one of the difficulties of the concept of modes in optical fibers, which understands what they are and how they differ from one another.

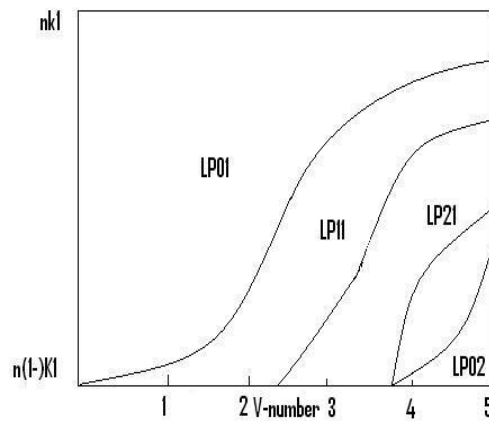
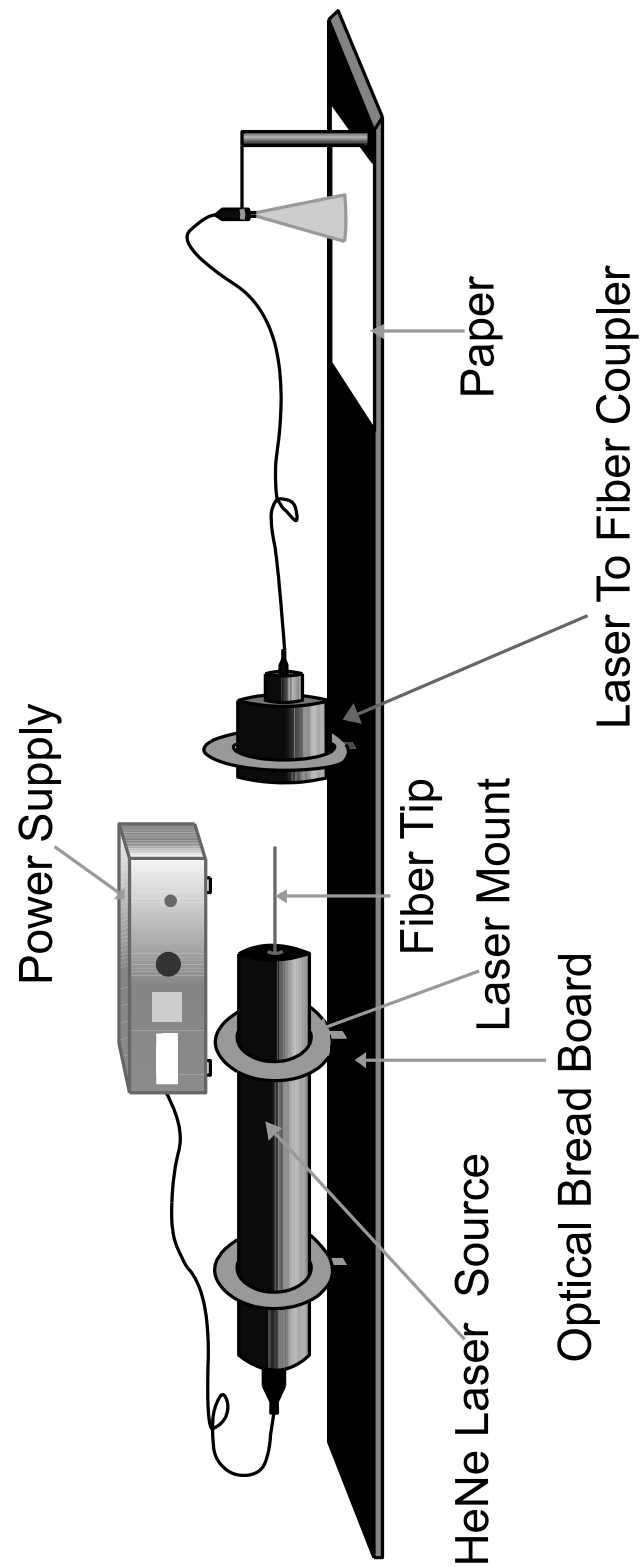


Figure 0.15. Low order linearly polarized modes of an optical fiber. Compare with Figure 0.13

0.3.2 Modes in Multimode Fibers

The multimode fibers used for telecommunications may have $a = 25 \mu\text{m}$ and $\text{NA} = 0.20$ or $a = 50 \mu\text{m}$ and $\text{NA} = 0.30$, so that for 633 nm light, the V – number will be about 50 or 150, respectively. This means that a large number of modes will be supported by the fiber. The amount of light carried by each mode will be determined by the input, or launch, conditions. For example, if the angular spread of the rays from the source is greater than the angular spread that can be accepted by the fiber (the NA of the input radiation is greater than the NA of the fiber) and the radius of the input beam is greater than the core radius of the fiber, then the fiber is said to be **overfilled (Fig. 0.16(a))**. That is, some of the light which the source will be putting into the fiber cannot be propagated by the fiber. Conversely, when the input beam NA is less than the fiber NA and the input beam radius is less than that of the fiber, the fiber is **underfilled (Fig. 0.16(a))** and only low-order modes (low-angle rays in the ray picture) will be excited in the fiber. These two distributions will yield different measures of attenuation, with the overfilled case having a higher loss than the underfilled case. In the ray picture, the higher-order rays will spend more of the time near the core-cladding interface and will have more of their evanescent field extending into the fiber cladding, resulting in higher attenuation. Also, if the fiber undergoes bending, the rays at high angle to the fiber axis may no longer satisfy the critical angle condition and not be totally internally reflected. Since power from these modes will radiate into the cladding and increase the attenuation, they are referred to as **radiation modes**. There is another class of modes called **leaky modes**. These modes have part of their electromagnetic energy distribution on the cladding, but none of the energy distribution is actually at the core-cladding interface. The energy in the core “leaks” into the cladding by a process known, from quantum mechanics, as tunneling. Leaky modes are not true guided modes, but may not be fully attenuated until the light has traveled long distances.



EXPERIMENT NO.1

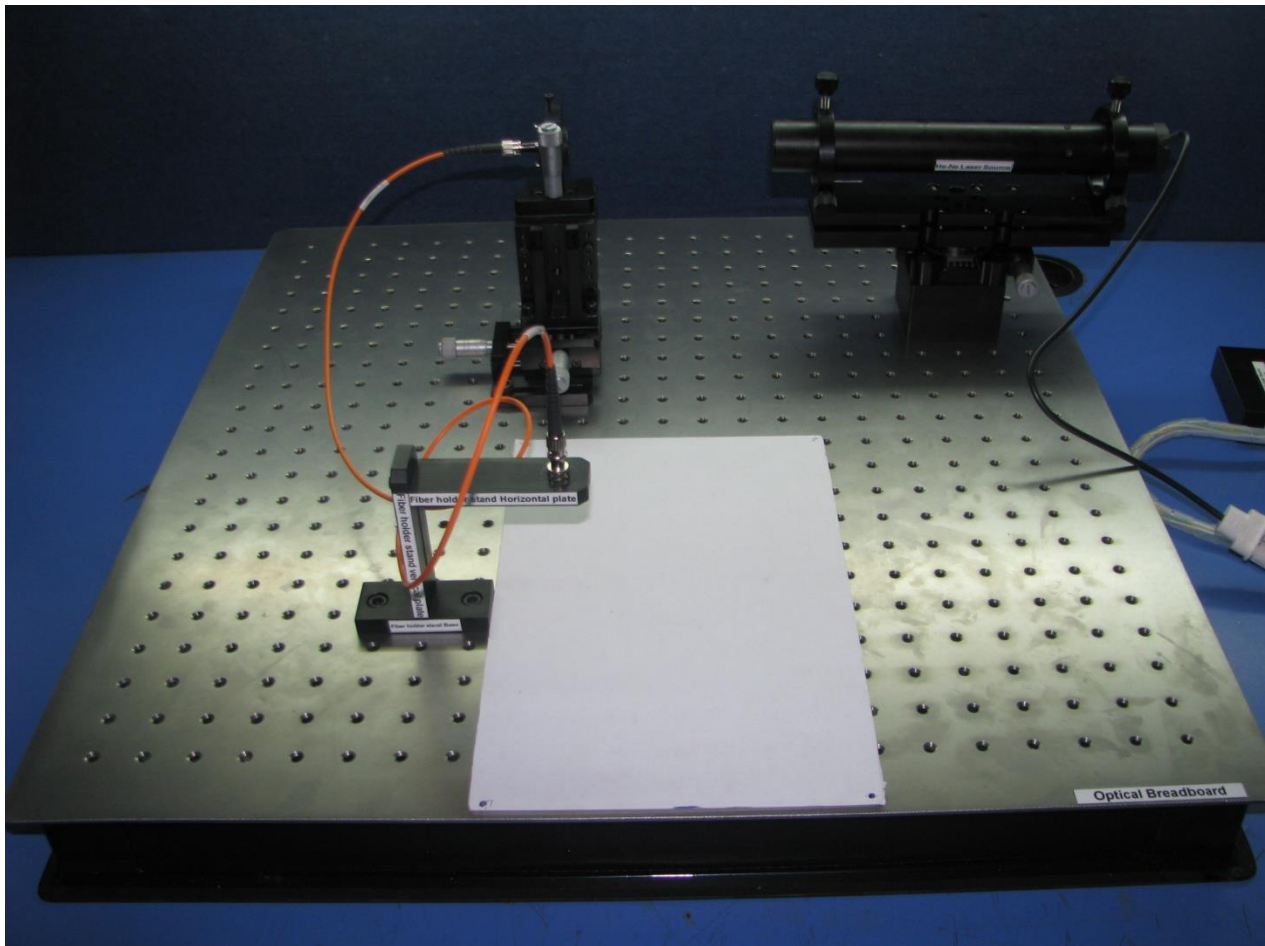


Fig. Experimental Setup for Mode Observations

EXPERIMENT NO. 1

NAME

Mode Observation in Single Mode & Multimode Optical Fiber

EQUIPMENTS

- 1) Optical breadboard.
- 2) He-Ne Laser Source.
- 3) Gimble mount - for Laser Head holder.
- 4) Gimble mount block.
- 5) X-Y-Z Positioner.
- 6) Laser to fiber coupler holder.
- 7) Laser to fiber coupler
- 8) 9 μm and 62.5 μm – single mode and multimode optical fiber patch cords.
- 9) Optical fiber holder stand.
- 10) Display screen.
- 11) Allen Bolts:-
 - a) M6 X 15mm – 2 Nos. for Gimble mount block.
 - b) M6 X 10mm – 2 Nos. for Gimble mount.
 - c) M6 X 10mm – 2 Nos. for X-Y-Z stage mounting.
 - d) M6 X 8mm – 2 Nos. for Laser to fiber coupler holder.
 - e) M3 X 8mm – 3 Nos. – Grub screws for Laser to fiber coupler holder.
 - f) M3 X 8mm – 3 Nos. – Grub screws for ST connector locking.
 - g) M4 X 10mm – 1 Nos. for fiber holder stands slider.
 - h) M6 X 10mm – 2 Nos. for fiber holder stand.
- 12) Allen Keys.

THEORY

Single Mode Fiber

In fiber-optic communication, a single-mode optical fiber (SMF) is an optical fiber designed to carry only a single ray of light (mode). This ray of light often contains a variety of different wavelengths. Although the ray travels parallel to the length of the fiber, it is often called the transverse mode since its electromagnetic vibrations occur perpendicular (transverse) to the length of the fiber. Single-mode optical fibers are also called monomode optical fibers, single-mode optical waveguides, or unimode fibers.

Unlike multi-mode optical fibers, single mode fibers do not exhibit dispersion resulting from multiple spatial modes. Single mode fibers are also better at retaining the fidelity of each light pulse over long distances than are multi-mode fibers. For these reasons, single-mode fibers can have a higher bandwidth than multi-mode fibers. Equipment for single mode fiber is more expensive than equipment for multi-mode optical fiber, but the single mode fiber itself is usually cheaper in bulk.

A typical single mode optical fiber has a core diameter between 8 and 10 μm [1] and a cladding diameter of 125 μm ; the wavelength of the light is 1310 or 1550 nm [2]. There are a number of special types of single-mode optical fiber which have been chemically or physically altered to give special properties, such as dispersion-shifted fiber and nonzero dispersion-shifted fiber. Data rates are limited by polarization mode dispersion and chromatic dispersion. In 2005, data rates of up to 10 gigabits per second were possible at distances of over 60 km with commercially available transceivers

The lowest order bound mode is ascertained for the wavelength of interest by solving Maxwell's equations for the boundary conditions imposed by the fiber, e.g., core (spot) size and the refractive indices of the core and cladding. The solution of Maxwell's equations for the lowest order bound mode will permit a pair of orthogonally polarized fields in the fiber, and this is the usual case in a communication fiber.

In step-index guides, single-mode operation occurs when the normalized frequency, V , is less than 2.405. For power-law profiles, single-mode operation occurs for a normalized frequency, V , less than approximately, where g is the profile parameter.

In practice, the orthogonal polarizations may not be associated with degenerate modes.

Multimode Fiber

Multi-mode optical fiber (multimode fiber or MM fiber or fibre) is a type of optical fiber mostly used for communication over shorter distances, such as within a building or on a campus. Multi-mode fibers support applications from 10 Mbit/s to 10 Gbit/s over link lengths of up to 550 meters—more than sufficient for the majority of premises applications.

The equipment used for communications over multi-mode optical fiber is less expensive than that for single-mode optical fiber. Typical transmission speeds/distances limits are 100 Mbit/s up to 2 km (100BASE-FX), 1 Gbit/s for distances up to 500–600 meters (1000BASE-SX), and 10 Gbit/s for distances up to 300 meters (10GBASE-SR).

Because of its high capacity and reliability, multi-mode optical fiber generally is used for backbone applications in buildings. An increasing number of users are taking the benefits of fiber closer to the user by running fiber to the desktop or to the zone. Standards-compliant architectures such as Centralized Cabling and Fiber to the Telecom Enclosure offer users the ability to leverage the distance capabilities of fiber by centralizing electronics in telecommunications rooms, rather than having active electronics on each floor.

Multimode fiber has higher "light-gathering" capacity than single-mode optical fiber. In practical terms, the larger core size simplifies connections and also allows the use of lower-cost electronics such as light-emitting diodes (LEDs) and vertical-cavity surface-emitting lasers (VCSELs) which operate at the 850 nm wavelength (single-mode fibers operate at 1310 or 1550 nm and require more expensive laser sources [1]). However, compared to single-mode fibers, the limit on speed $\times$ distance is lower. Because multi-mode fiber has a larger numerical aperture than single-mode fiber, it supports more than one propagation mode, hence it is limited by modal dispersion, where single mode is not. Consequently, multi-mode fiber has higher pulse spreading rates than single mode fiber, limiting multi-mode fiber's information transmission capacity.

Multi-mode fibers are described by their core and cladding diameters. Thus, 62.5/125 μm multimode fiber has a core size of 62.5 micrometers (μm) and a cladding diameter of 125 μm . In addition, multi-mode fibers are described using a system of classification determined by the ISO 11801 standard — OM1, OM2, and OM3 — which is based on the bandwidth of the multi-mode fiber.

For many years 62.5/125 μm (OM1) and conventional 50/125 μm multi-mode fiber (OM2) were widely deployed in premises applications. These fibers easily support applications ranging from Ethernet (10 Mbit/s) to Gigabit Ethernet (1 Gbit/s) and, because of their relatively large core size, were ideal for use with LED transmitters. Newer deployments often use laser-optimized 50/125 μm multi-mode fiber (OM3). Fibers that meet this designation provide sufficient bandwidth to support 10 Gigabit Ethernet up to 300 meters. Optical fiber manufacturers have greatly refined their manufacturing process since that standard was issued and cables can be made that support 10 GbE up to 550 meters. Laser optimized multi-mode fiber (LOMMF) is designed for use with 850 nm VCSELs.

The migration to LOMMF/OM3 has occurred as users upgrade to higher speed networks. LEDs have a maximum modulation rate of 622 Mbit/s because they cannot be turned on/off fast enough to support higher bandwidth applications. VCSELs are capable of modulation over 10 Gbit/s and are used in many high speed networks.

VCSEL power profiles, along with variations in fiber uniformity, can cause modal dispersion which is measured by differential modal delay (DMD). Modal dispersion is an effect that caused by the different speeds of the individual modes in a light pulse. The net effect causes the light pulse to separate or spread over distance, making it difficult for receivers to identify the individual 1's and 0's. The greater the length, the greater the modal dispersion. To combat modal dispersion, LOMMF is manufactured in a way that eliminates variations in the fiber which could affect the speed that a light pulse can travel. The refractive index profile is enhanced for VCSEL transmission and to prevent the pulse spreading. As a result the fibers maintain signal integrity over longer distances, thereby maximizing bandwidth.

PROCEDURE

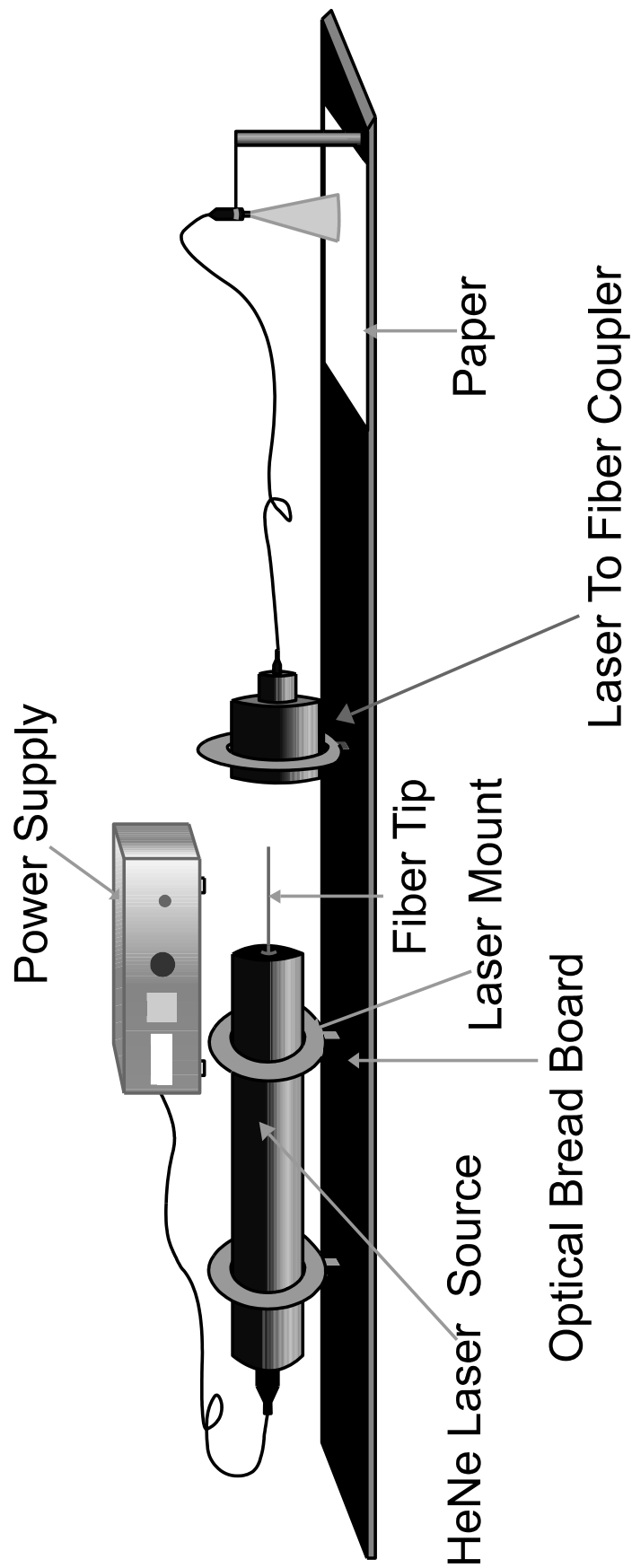
- Identify the mechanical components, Laser source, Laser to fiber coupler and optical fibers mentioned above from the kit.
- Refer to the placement diagram 1 for mounting of the mechanical components.
- First fix the Gimble mount block on the optical bread board using M6 X 15 mm Allen bolts at the position shown in diagram 1.
- Fix the Gimble mount on the Gimble mount block using M6 X 10 mm Allen bolts.
- Remove the top half of the Gimble mount and insert the He-Ne laser head with packing sleeve in the Gimble mount. Fix the top half of the mount to hold the laser tightly. Do not over tighten, as this will damage the laser. Gimble mount has θ and ϕ degrees of freedom, which can be adjusted by the screws provided on the mount.
- Now fix the X-Y-Z mount on optical breadboard in front of laser head using M6 X 10 mm Allen bolts. Also mount the laser to fiber coupler holder on X-Y-Z platform with the direction shown in the diagram 2 using M6 X 5mm Allen bolts.
- Align the center of the laser to fiber coupler holder to the laser beam. For this power on the laser source.
- Insert the laser to fiber coupler in the coupler holder mounted on X-Y-Z platform with ST receptacle facing opposite to laser head. Fix the coupler with M3 X 8mm grub screws.
- Now adjust the X-Y-Z stage and Gimble mount to align the laser beam to fall on the center lens of the laser to fiber coupler. One should observe clean circular spot of laser beam coming out of the ST receptacle. Use white screen for observation, which

provided with the kit. Here the back reflection from the Laser to fiber coupler lens should go back into the exit aperture of the laser head i.e. in the direction of the incident beam. This is very important step in alignment

- Now select 62.5 μm multimode patch cord - (Orange Colored) Insert one end of the patch cord into the ST receptacle of the laser to fiber coupler.
- Mount the fiber holder stand on the optical breadboard as shown in diagram 3 using M6 X 10 mm Allen bolt. This holder has ST receptacle attached to it.
- Insert the other end of the patch cord into the ST receptacle of the fiber holder stand.
- Put the white screen below the fiber holder stand to observe the light coming out of the patch cord.
- Now Refer to the "LASER TO FIBER SOURCE COUPLERS APPLICATION NOTE" document attached at the back of the manual. Refer to the diagram and description under the title "RECEPTACLE STYLE PHYSICAL CONTACT SOURCE COUPLERS (HUC MODEL)" on the 1st page of the application notes.
- As one can see the coupler has two main parts LASER HEAD DISK ADAPTOR and CONNECTOR RECEPTACLE WITH FOCUSING LENS. These two parts are separated by O-RING and connected together with three 80 TPI LOCKING SCREWS. Also are provided three 80 TPI TILT ADJUSTMENT SCREWS.
- After following the steps up to 13th one, one should focus on the adjustment of the TILT ADJUSTMENT SCREWS. Before that 1st slightly loosen the LOCKING SCREWS and then start adjusting TILT ADJUSTMENT SCREWS with the help of special Allen screwdriver provided with the kit.
- While patiently working with the alignment procedure one can observe bright red multimode pattern on the screen kept below fiber holder stand.
- Now to observe the modes from the single mode fiber just replace the multimode patch cord with single mode one without disturbing any of the previous alignments or settings.
- Again use the TILT ADJUSTMENT SCREWS to do the fine alignment to observe the modes propagating through single mode fiber.
- When we adjust the TILT ADJUSTMENT SCREWS, we change the angle of incident of the light at the input surface of the fiber tip; this causes the observation of various modes in optical fiber.

EXPERIMENT NO.2

Experimental Setup For Numerical Aperture



EXPERIMENT NO 2

NAME

Finding the Numerical Aperture of the Multimode optical fiber

EQUIPMENTS

- 1) Optical breadboard.
- 2) He-Ne Laser Source.
- 3) Gimble mount - for Laser Head holder.
- 4) Gimble mount block.
- 5) Rotary stage
- 6) Rotary stage mount block.
- 7) Rotary stage fiber holder with receptacle.
- 8) Photo detector cap with receptacle.
- 9) Laser Power Meter (Read out unit, Photo detector head and connecting cable)
- 10) 62.5/125 μm multimode optical fiber patch cord
- 11) 100/140 μm multimode optical fiber patch cord
- 12) Allen Bolts:-
 - a) M6 X 15mm – 2 Nos. for Gimble mount block.
 - b) M6 X 10mm – 2 Nos. for Gimble mount.
 - c) M6 X 15mm – 2 Nos. for Rotary stage mount block.
 - d) M6 X 15mm – 2 Nos. for Rotary stage.
- 12) Allen Keys.

THEORY

In laser physics, the numerical aperture is defined slightly differently. Laser beams spread out as they propagate, but slowly. Far away from the narrowest part of the beam, the spread is roughly linear with distance—the laser beam forms a cone of light in the "far field". The same relation gives the NA,

$$\text{NA} = n \sin \theta$$

Multimode optical fiber will only propagate light that enters the fiber within a certain cone, known as the acceptance cone of the fiber. The half-angle of this cone is called the acceptance angle, θ_{max} . For step-index multimode fiber, the acceptance angle is determined only by

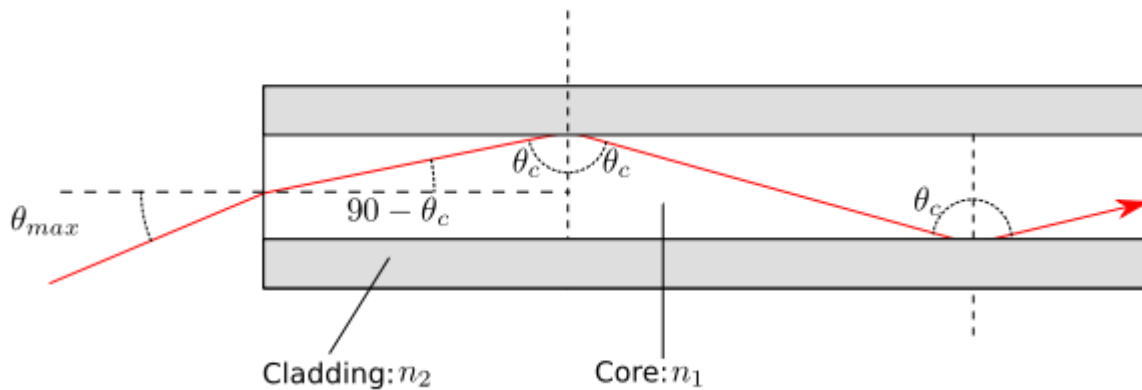
the indices of refraction:

$$n \sin \theta_{\text{max}} = \sqrt{n_1^2 - n_2^2},$$

where n_1 is the refractive index of the fiber core, and n_2 is the refractive index of the cladding.



Fig. Experimental setup for finding Numerical Aperture



When a light ray is incident from a medium of refractive index n to the core of index n_1 , Snell's law at medium-core interface gives

$$n \sin \theta_i = n_1 \sin \theta_r.$$

From the above figure and using trigonometry, we get :

$$\sin \theta_r = \sin \theta_2 = \sin (90^\circ - \theta_c) = \cos \theta_c \text{ Where}$$

$$\theta_c = \sin^{-1} \frac{n_2}{n_1} \text{ is the critical angle for total internal reflection, since}$$

Substituting for $\sin \theta_r$ in Snell's law we get:

$$\frac{n}{n_1} \sin \theta_i = \cos \theta_c.$$

By squaring both sides

$$\frac{n^2}{n_1^2} \sin^2 \theta_i = \cos^2 \theta_c = 1 - \sin^2 \theta_c = 1 - \frac{n_2^2}{n_1^2}.$$

Thus,

$$n \sin \theta_i = \sqrt{n_1^2 - n_2^2},$$

from where the formula given above follows.

This has the same form as the numerical aperture in other optical systems, so it has become common to define the NA of any type of fiber to be

$$NA = \sqrt{n_1^2 - n_2^2},$$

where n_1 is the refractive index along the central axis of the fiber. Note that when this definition is used, the connection between the NA and the acceptance angle of the fiber becomes only an approximation. In particular, manufacturers often quote “NA” for single-mode fiber based on this formula, even though the acceptance angle for single-mode fiber is quite different and cannot be determined from the indices of refraction alone.

The number of bound modes, the mode volume, is related to the normalized frequency and thus to the NA.

In multimode fibers, the term equilibrium numerical aperture is sometimes used. This refers to the numerical aperture with respect to the extreme exit angle of a ray emerging from a fiber in which equilibrium mode distribution has been established.

PROCEDURE

- 1) Identify the mechanical components, Laser source, optical fiber and Laser power meter mentioned above from the kit.
- 2) Refer to the placement diagram 4 for mounting of the mechanical components.
- 3) First fix the Gimble mount block on the optical breadboard using M6 X 15mm Allen bolts I position shown in diagram 4.
- 4) Fix the Gimble mount on the Gimble mount block using M6 X 10 mm Allenbolts.
- 5) Remove the top half of the Gimble mount and insert the He-Ne laser head with packing sleeve in the Gimble mount. Fix the top half of the mount to hold the laser tightly. Do not over tighten, as this will damage the laser. Gimble mount has θ and ϕ degrees of freedom, which can be adjusted by the screws provided on the mount.
- 6) Now fix the rotary stage mount block on optical breadboard in front of laser head using M6 X 15 mm Allen bolts. Also mount the rotary stage on this block using M6 X 15mm Allen bolts.
- 7) Fix the rotary stage fiber holder with receptacle on the rotary stage at its center hole with the receptacle facing opposite to the laser head.
- 8) Align the center of the receptacle to the laser beam. For this power on the laser source.
- 9) Now select 62.5/125- μm or 100/140 μm multimode patch cord. Insert one end of the patch cord into the ST receptacle of the rotary stage fiber holder.
- 10) Fix the photo detector cap with receptacle to the photo detector head of power meter. Now insert the other end of the patch cord into the ST receptacle of the Photo detector cap.
- 11) Connect the readout of the laser power meter to detector head with its connecting cable and power on the readout. To get the stable readings, power meter is required to be worm up for minimum of 15 min.
- 12) Loosen the locking screw provided in front of the rotary stage and set the micrometer to 5 mm of its linear scale. I.e the central position.
- 13) Now with fine adjustment of the Gimble mount screws, direct the laser beam spot on the fiber face such that the back reflection from it should go back in the exit of the laser beam. If required rotate the fiber holder manually-i.e. Not with the micrometer.
- 14) After this lock the rotary stage by tightening the locking screw provided in front of the rotary stage. Now set the zero marking on the circular dial to zero of angular scale of

rotary stage. Locking the rotary stage by locking screw, allows it to rotate by micrometer.

- 15) One has to rotate the rotary stage on either side-using micrometer to find out maximum reading on laser power meter. The point, at which you get the maximum reading, is the stage of exact alignment of laser beam to the face of the fiber tip. This point is to be considered as zero for micrometer setting.
- 16) Loosen the locking screw provided in front of the rotary stage and set the micrometer to 0 mm of its linear scale.
- 17) Now rotate the rotary stage by micrometer in clockwise direction with the step of one degree and note the reading on laser power meter. Continue measurements till you reach 8 degrees in clockwise direction.
- 18) To take the readings in anticlockwise direction, first reach to the maximum reading on laser power meter. Then loosen the locking screw provided in front of the rotary stage and set the micrometer to 10 mm of its linear scale. After this lock the rotary stage by tightening the locking screw provided in front of the rotary stage.
- 19) Now rotate the rotary stage by micrometer in anticlockwise direction with the step of one degree and note the reading on laser power meter. Continue measurements till you reach 8 degrees in anticlockwise direction.
- 20) Plot the graph of intensity of the laser beam verses sine of θ , while doing this normalize the reading at maximum to 100 %. Refer to the graph provided in fig 5.
- 21) The NA is determined using one half of the full angle between the two 5% intensity points on the graph.

OBSERVATION

Finding the Numerical Aperture of the Multimode optical fiber.

Observation:- 1) Anticlockwise Direction (+ve).

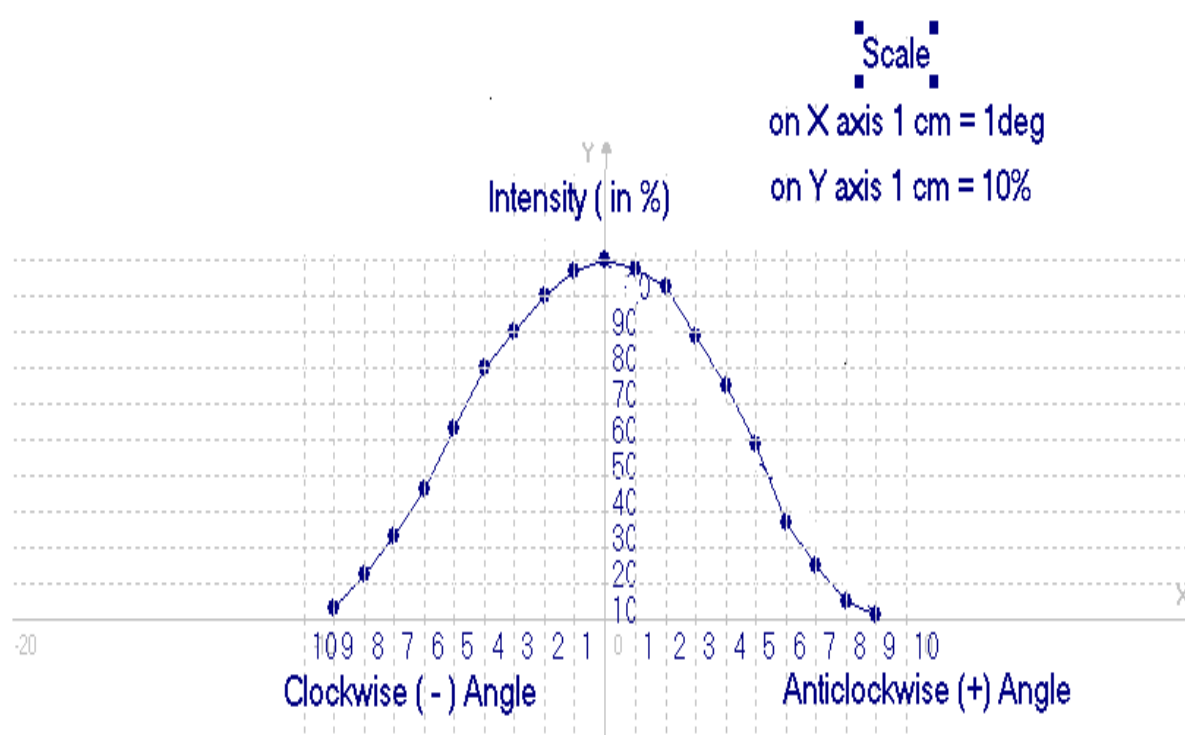
Sr.no.	Angle (degree)	Laser intensity (uW)	In (%)
1	0	54.8	100
2	1	53.7	98
3	2	51.0	93
4	3	43.5	79
5	4	35.5	65
6	5	26.6	49
7	6	15.0	27
8	7	8.0	15
9	8	2.9	5
10	9	0.6	1

2) Clockwise Direction (-ve).

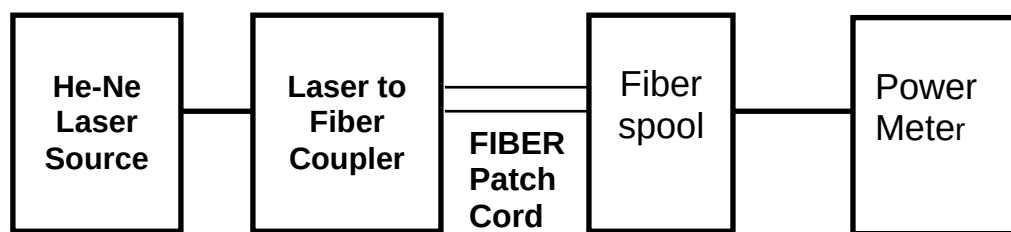
Sr.no.	Angle (degree)	Laser intensity (uW)	In (%)
1	0	54.8	100
2	1	53.3	97

3	2	49.5	90
4	3	43.9	80
5	4	38.1	70
6	5	29.0	53
7	6	19.9	36
8	7	12.6	23
9	8	6.8	12
10	9	1.4	3

$$NA = \sin \theta.$$



EXPERIMENT NO.3



CUTBACK METHOD

EXPERIMENT NO.3

NAME

Propagation Loss in Multimode optical fiber using cutback method

EQUIPMENTS

- 1) Optical breadboard.
 - 2) He-Ne Laser Source.
 - 3) Gimble mount – for Laser Head holder.
 - 4) Gimble mount block.
 - 5) X-Y-Z Positioner.
 - 6) Laser to fiber coupler holder.
 - 7) Laser to fiber coupler
 - 8) 62.5/125 μm multimode optical fiber patch cords
 - 9) Photo detector cap with receptacle.
 - 10) Laser Power Meter (Read out unit, Photo detector head and connecting cable)
 - 11) ST to ST receptacle
 - 12) FOL-Fiber containing 100 meter, 500 meter and 1000 meter multimode fiber cables.
 - 13) Allen Bolts:-
 - a) M6 X 15mm – 2 Nos. for Gimble mount block.
 - b) M6 X 10mm – 2 Nos. for Gimble mount.
 - c) M6 X 10mm – 2 Nos. for X-Y-Z stage mounting.
 - d) M6 X 5mm – 2 Nos. for Laser to fiber coupler holder.
 - e) M3 X 8mm – 3 Nos. – Grub screws for Laser to fiber coupler
- Holder
- 14) Allen Keys.

THEORY

Cutback Method

1. performing the desired measurements on a long length of the fiber under test,
2. cutting the fiber under test at a point near the launching end,
3. repeating the measurements on the short length of fiber, and
4. subtracting the results obtained on the short length to determine the results for the residual long length.

The benefit of this technique is that it allows measurement of the fiber characteristics without introducing errors due to variation in the launch conditions. For example, the coupling efficiency of the light source is kept consistent between the initial and the cutback measurements



Fig. Experimental Setup for finding Propagation Loss

The cut should be made to retain 1 meter or more of the fiber, in order to establish equilibrium mode distribution conditions for the second measurement. In a multimode fiber, the lack of an equilibrium mode distribution could introduce errors in the measurement due to output coupling effects. In a single-mode fiber, measuring a shorter cutback fiber could result in significant transmission of cladding modes (light carried in the cladding rather than the core of the optical fiber), distorting the measurement. The errors introduced will result in conservative results (*i.e.*, higher transmission losses and lower bandwidths) than would be realized under equilibrium conditions.

Several characteristics may be determined using the same test fiber.

A variation of the cutback technique is the substitution method, in which measurements are made on a full length of fiber, and then on a short length of fiber having the same characteristics (core size, numerical aperture), with the results from the short length being subtracted to give the results for the full length.

PROCEDURE

- Identify the mechanical components, Laser source, and optical fiber, Laser to fiber coupler and Laser power meter mentioned above from the kit.
- Refer to the placement diagram 6 for mounting of the mechanical components.

- First fix the Gimble mount block on the optical bread board using M6 X 15 mm Allen bolts at the position shown in diagram 6.
- Fix the Gimble mount on the Gimble mount block using M6 X 10 mm Allen bolts.
- Remove the top half of the Gimble mount and insert the He-Ne laser head with packing sleeve in the Gimble mount. Fix the top half of the mount to hold the laser tightly. Do not over tighten, as this will damage the laser. Gimble mount has θ and Φ degrees of freedom, which can be adjusted by the screws provided on the mount.
- Now fix the X-Y-Z mount on optical breadboard in front of laser head using M6 X 10 mm Allen bolts. Also mount the laser to fiber coupler holder on X-Y-Z platform with the direction shown in the diagram 7 using M6 X 5mm Allen bolts.
- Align the center of the laser to fiber coupler holder to the laser beam. For this power on the laser source.
- Insert the laser to fiber coupler in the coupler holder mounted on X-Y-Z platform with ST receptacle facing opposite to laser head. Fix the coupler with M3 X 8mm grub screws.
- Now adjust the X-Y-Z stage and Gimble mount to align the laser beam to fall on the center lens of the laser to fiber coupler. One should observe clean circular spot of laser beam coming out of the ST receptacle. Use white screen for observation, which provided with the kit. Here the back reflection from the Laser to fiber coupler lens should go back into the exit aperture of the laser head i.e. in the direction of the incident beam. This is very important step in alignment.
- Now select 62.5 μm multimode patch cord. Insert one end of the patch cord into the ST receptacle of the laser to fiber coupler.
- Fix the photo detector cap with receptacle to the photo detector head of laser power meter. Now insert the other end of the patch cord into the ST receptacle of the Photo detector cap.
- Connect the readout of the laser power meter to detector head with its connecting cable and power on the readout. To get the stable readings, power meter is required to be warm up for minimum of 15 min.
- Now Refer to the "LASER TO FIBER SOURCE COUPLERS APPLICATION NOTE" document attached at the back of the manual. Refer to the diagram and description under the title "RECEPTACLE STYLE PHYSICAL CONTACT SOURCE COUPLERS (HUC MODEL)" on the 1st page of the application notes.
- As one can see the coupler has two main parts LASER HEAD DISK ADAPTOR and CONNECTOR RECEPTACLE WITH FOCUSING LENS. These two parts are separated by O-RING and connected together with three 80 TPI LOCKING SCREWS. Also are provided three 80 TPI TILT ADJUSTMENT SCREWS.
- After following the steps up to 12th one, one should focus on the adjustment of the TILT ADJUSTMENT SCREWS. Before that 1st slightly loosen the LOCKING SCREWS and then start adjusting TILT ADJUSTMENT SCREWS with the help of special Allen screwdriver provided with the kit.
- At this stage observe the readings on the power meter and try to adjust the TILT ADJUSTMENT SCREWS to get the maximum reading. This ensures that maximum intensity is coupled to the optical fiber.
- Mark this reading as P_i the input power that we will launch in to the fiber cable of different lengths.
- Now remove the patch cord from detector head only and using ST to ST

Receptacle, connect it to one end of 100 meter of fiber cable from FOM 3A model. The other end of the 100 meter fiber cable is to be connected to power meter head using another 62.5 μm multimode patch cord.

- Read the value on the power meter and mark it as power P_o . With the standard formula for dB calculate the power, which is the propagation loss in 100 meter cable. But this value will also involve coupling losses at various stages.

OBSERVATION

- 1) P_i = reading through 1mtr 62.5/125 micron multimode patch cord.
 $P_i = 0.985 \text{ mW}$.
- 2) $P_{1\text{out}}$ (fiber 1 = 100 meter) = 0.701 mW.
- 3) $P_{2\text{out}}$ (fiber 2 = 500 meter) = 0.216 mW.
- 4) $P_{3\text{out}}$ (fiber 3 = 1000 meter) = 0.064 mW.

EXPERIMENT NO.4



Coupling Efficient Of A Laser To Fiber Coupler

EXPERIMENT NO.4

NAME

Measurement of coupling efficiency of a Laser to Fiber Coupler (LTFC)

EQUIPMENTS

- 1) Optical breadboard.
- 2) He-Ne Laser Source.
- 3) Gimble mount – for Laser Head holder.
- 4) Gimble mount block.
- 5) Laser to fiber coupler with V groove fiber holder & X-Y-Z stage.
- 6) 62.5 μm multimode optical fiber pigtail
- 7) Photo detector cap with receptacle.
- 8) Laser Power Meter (Read out unit, Photo detector head and connecting cable)
- 9) Allen Bolts:-
 - a) M6 X 10mm – 2 Nos. for Gimble mount block.
 - b) M6 X 15mm – 2 Nos. for Gimble mount.
 - c) M6 X 10mm – 2 Nos. for Laser to fiber coupler block.
 - d) M6 X 15mm – 2 Nos. for Laser to fiber coupler.
- 10) Allen Keys.

THEORY

Coupling loss in fiber optics refers to the power loss that occurs when coupling light from one optical device or medium to another in optical systems, where there is a change in index of refraction (most commonly at a fiber/air interface), a portion of the energy is reflected back into the source component. Another major source of optical coupling loss is geometrical. As an example, two fibers coupled end-to-end may not be precisely aligned, with the result that the two cores overlap somewhat. Light exiting the source fiber at a portion of its core that is not aligned with the core of the receiving fiber will not (in general) be coupled into the second fiber. [Note that while some such light will be coupled into the second fiber, it is not likely to be efficiently coupled, nor will it generally travel in an appropriate mode in the second fiber.] Similarly, even for two perfectly aligned cores, where there is a gap of any significant distance between the two fibers, there will be some geometric loss due to spread of the beam. Some percentage of the light rays exiting the source fiber face will not intersect the second fiber within its entrance cone.

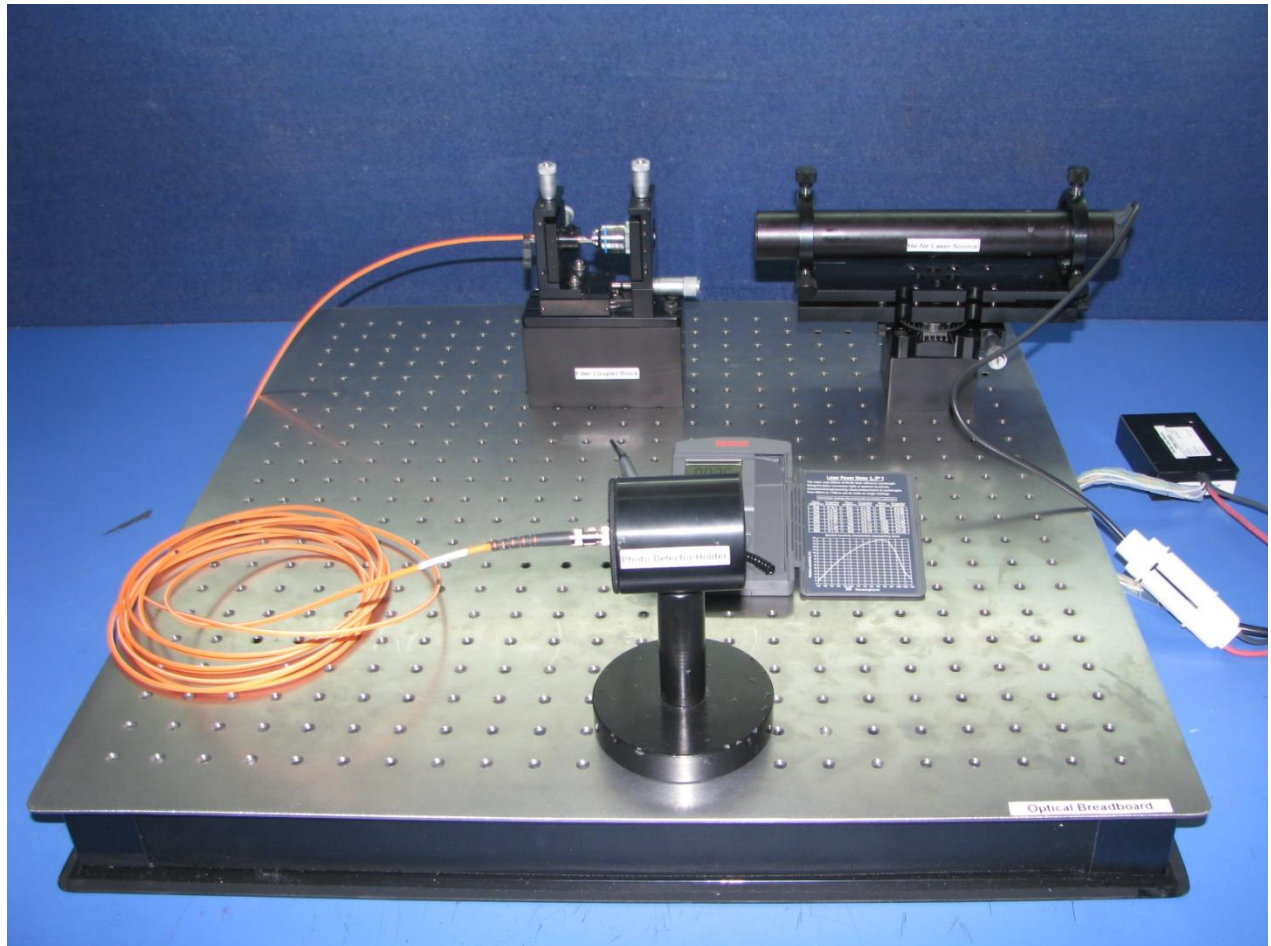
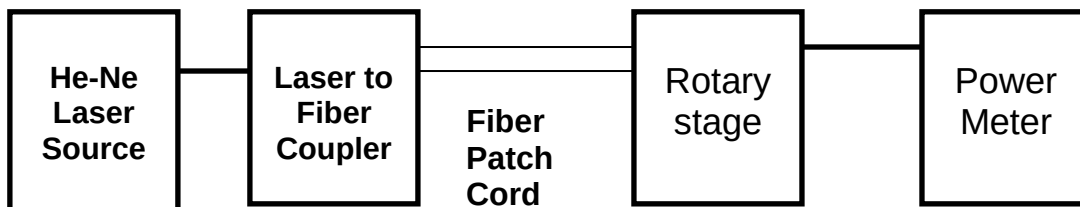


Fig. Experimental Setup for Measuring Coupling Efficiency

PROCEDURE

- Identify the mechanical components, Laser source, optical fiber and Laser power meter mentioned above from the kit.
- Refer to the placement diagram 11 for mounting of the mechanical components.
- First fix the Gimble mount block on the optical bread board using M6 X 15 mm Allen bolts at the position shown in diagram 11.
- Fix the Gimble mount on the Gimble mount block using M6 X 10 mm Allen bolts.
- Remove the top half of the Gimble mount and insert the He-Ne laser head with packing sleeve in the Gimble mount. Fix the top half of the mount to hold the laser tightly. Do not over tighten, as this will damage the laser. Gimble mount has θ and Φ degrees of freedom, which can be adjusted by the screws provided on the mount.
- Keep the detector head in front of the laser source.
- Now adjust the Gimble mount to align the laser beam to fall on the center of the photo detector to get the maximum reading on power meter readout. For this power on the laser source.
- Connect the readout of the laser power meter to detector head with its connecting cable and power on the readout. To get the stable readings, power meter is required to be warm up for minimum of 15 min.
- Take a reading and refer it as a power P1.
- Now refer to the placement diagram 12 for mounting of the Laser to fiber coupler.
- Fix the Laser to fiber coupler block on optical breadboard in front of laser head using M6 X 10 mm Allen bolts. Also mount the Laser to fiber coupler on this block using M6 X 15mm Allen bolts.
- Now adjust the Gimble mount and laser to fiber coupler to align the laser beam to fall on the center of the laser to fiber coupler. One should observe clean circular spot of laser beam coming out of the laser to fiber coupler. Use white screen for observation, which is provided with the kit. Here the back reflection from the Laser to fiber coupler should go back into the exit aperture of the laser head i.e. in the direction of the incident beam. This is very important step in alignment.
- Take 62.5-micron multimode optical fiber pigtail (orange colored). With the help of 3 in 1 fiber stripper remove around 6 cm of orange jacket of the fiber from open-end side. Cut the Kevlar members and you should have fiber with buffer left. With the same stripper remove around 1.5 cm of buffer on the fiber, after this you will have bare fiber, which needs to be cleaved using Carbide Fiber Scribe. Cleave the bare fiber exact vertically at 0.5 cm from buffer.
- Place the buffered fiber in a V-groove assembly and lock it from jacket end using locking sleeve. Now insert this assembly in a laser to fiber coupler with bare end of the fiber facing lens assembly of the coupler and fix this end with another locking sleeve. Align the bare fiber tip at the center of the focal spot of the lens assembly.
- Fix the photo detector cap with receptacle to the photo detector head of laser power meter. Now insert the other end of the pigtail into the ST receptacle of the Photo detector cap.
- One has to adjust the X-Y scale of Laser to fiber coupler using thumbscrew to get the maximum reading on laser power meter. Note down the reading as a power P2.
- Now find out the coupling efficiency of a Laser to Fiber Coupler using formula $(P2/P1) * 100$ to get it in %.

EXPERIMENT NO.5



Measurement Of The Mode Field Diameter Of AnOptical Fiber

EXPERIMENT NO. 5

NAME

Mode Field Diameter of Multimode & Single mode optical fiber

OBJECTIVE

Measurement of Mode Field Diameter of Multimode & Single mode optical fiber

EQUIPMENTS

- 1) Optical breadboard.
- 2) He-Ne Laser Source.
- 3) Gimble mount – for Laser Head holder.
- 4) Gimble mount block.
- 5) X-Y-Z Positioner.
- 6) Laser to fiber coupler holder.
- 7) Laser to fiber coupler
- 8) Rotary stage
- 9) Rotary stage mount block.
- 10) Rotary stage fiber holder with receptacle.
- 11) Photo detector cap with knife edge slit.
- 12) Laser Power Meter (Read out unit, Photo detector head and connecting cable)
- 13) 9 μm , 62.5 μm single and multimode optical fiber patch cords.
- 14) Allen Bolts:-
 - a) M6 X 15mm – 2 Nos. for Gimble mount block.
 - b) M6 X 10mm – 2 Nos. for Gimble mount.
 - c) M6 X 10mm – 2 Nos. for X-Y-Z stage mounting.
 - d) M6 X 5mm – 2 Nos. for Laser to fiber coupler holder.
 - e) M3 X 8mm – 3 Nos. – Grub screws for Laser to fiber coupler holder.
 - f) M6 X 15mm – 2 Nos. for Rotary stage mount block.
 - g) M6 X 15mm – 2 Nos. for Rotary stage.
- 15) Allen Keys.

THEORY

In fiber optics, the mode field diameter (MFD) is an expression of distribution of the irradiance, *i.e.*, the optical power per unit area, across the end face of a single-mode fiber.

For a Gaussian power distribution in a single-mode optical fiber, the mode field diameter is that at which the electric and magnetic field strengths are reduced to $1/e$ of their maximum values, *i.e.*, the diameter at which power is reduced to $1/e^2$ of the maximum power, because the power is proportional to the square of the field strength.

Irradiance	E	watt per square meter	$\text{W} \cdot \text{m}^{-2}$	power incident on a surface. Sometimes confusingly called "intensity"
------------	-----	-----------------------	--------------------------------	--

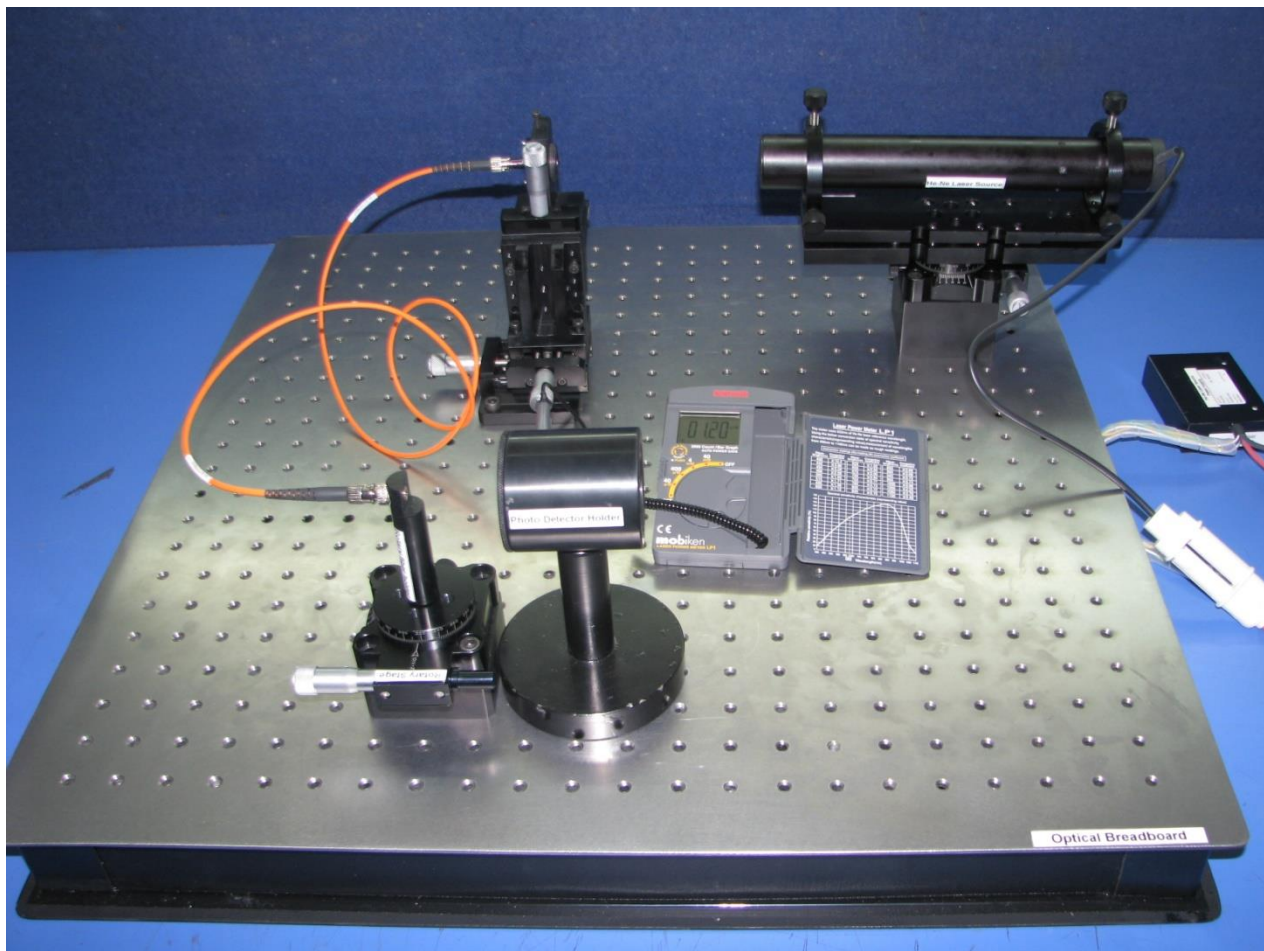


Fig. Experimental Setup for Finding Mode Field Diameter

PROCEDURE

- Identify the mechanical components, Laser source, Laser to fiber coupler, optical fibers and Laser power meter mentioned above from the kit.
- Refer to the placement diagram 8 for mounting of the mechanical components.
- First fix the Gimble mount block on the optical breadboard using M6 X 15mm Allen bolts at the position shown in diagram 8.
- Fix the Gimble mount on the Gimble mount block using M6 X 10 mm Allen bolts.
- Remove the top half of the Gimble mount and insert the He-Ne laser head with
 - Packing sleeve in the Gimble mount. Fix the top half of the mount to hold the laser tightly. Do not over tighten, as this will damage the laser. Gimble mount has θ and Φ degrees of freedom, which can be adjusted by the screws provided on the mount.
- Now fix the X-Y-Z mount on optical breadboard in front of laser head using
 - M6 X 10 mm Allen bolts. Also mount the laser to fiber coupler holder on
 - X-Y-Z platform with the direction shown in the diagram 9 using M6 X 5mm
 - Allen bolts.
- Align the center of the laser to fiber coupler holder to the laser beam. For this power on the laser source.
- Insert the laser to fiber coupler in the coupler holder mounted on X-Y-Z platform with ST receptacle facing opposite to laser head. Fix the coupler with M3 X 8mm grub screws.
- Now adjust the X-Y-Z stage and Gimble mount to align the laser beam to fall on the center lens of the laser to fiber coupler. One should observe clean circular spot of laser beam coming out of the ST receptacle. Use white screen for observation, which provided with the kit. Here the back reflection from the Laser to fiber coupler lens should go back into the exit aperture of the laser head i.e. in the direction of the incident beam. This is very important step in alignment.
- Now select 62.5/125 μm multimode patch cord. Insert one end of the patch cord into the ST receptacle of the laser to fiber coupler.
- As shown in the diagram 9, fix the rotary stage mount block on optical breadboard using M6 X 15 mm Allen bolts. Also mount the rotary stage on this block using M6 X 15mm Allen bolts.
- Fix the rotary stage fiber holder with receptacle on the rotary stage at its center hole with the receptacle facing opposite to the breadboard area.
- Insert other end of the patch cord into the ST receptacle of the rotary stage fiber holder.
- Fix the photo detector cap with knife edge slit to the photo detector head of power meter.
- Connect the readout of the laser power meter to detector head with its connecting cable and power on the readout. To get the stable readings, power meter is required to be warm up for minimum of 15 min.
- By this time with laser source turned on, you will see bright red circular spot showing multiple modes propagating at the end of the patch cord connected to the rotary stage fiber holder. Align the center of this spot to the middle of the knife edge.
- Loosen the locking screw provided in front of the rotary stage and set the micrometer to 5 mm of its linear scale. I.e the central position.

- After this lock the rotary stage by tightening the locking screw provided in front of the rotary stage. Now set the zero marking on the circular dial to zero of angular scale of rotary stage. Locking the rotary stage by locking screw, allows it to rotate by micrometer.
- One has to rotate the rotary stage on either side-using micrometer to find out maximum reading on laser power meter. The point, at which you get the maximum reading, is to be considered as zero for micrometer setting.
- Loosen the locking screw provided in front of the rotary stage and set the micrometer to 0 mm of its linear scale.
- Now rotate the rotary stage by micrometer in clockwise direction with the step of one degree and note the reading on laser power meter. Continue measurements till you reach 8 degrees in clockwise direction.
- To take the readings in anticlockwise direction, first reach to the maximum reading on laser power meter. Then loosen the locking screw provided in front of the rotary stage and set the micrometer to 10 mm of its linear scale. After this lock the rotary stage by tightening the locking screw provided in front of the rotary stage.
- Now rotate the rotary stage by micrometer in anticlockwise direction with the step of one degree and note the reading on laser power meter. Continue measurements till you reach 8 degrees in anticlockwise direction.

OBSERVATION

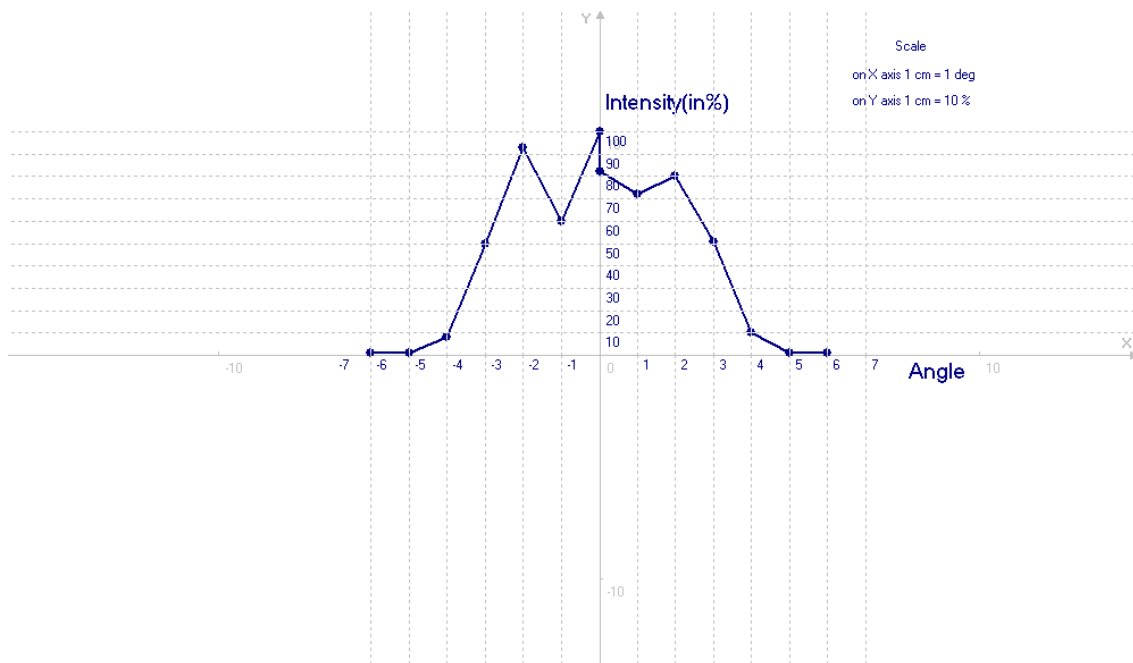
0) 9/125 micron Single Mode fibercable :-

Observation:- 1) Anticlockwise Direction (+ve).

Sr.no.	Angle (degree)	Laser intensity (μW)	In (%)
1	0	20	82
2	1	17.6	72
3	2	19.6	80
4	3	12.5	51
5	4	2.4	10
6	5	0.4	1
7	6	0.3	1

2) Clockwise Direction (-ve).

Sr.no.	Angle (degree)	Laser intensity (μW)	In (%)
1	0	24.5	100
2	1	14.6	60
3	2	22.8	93
4	3	12.2	50
5	4	2.0	8
6	5	0.3	1
7	6	0.3	1



B) 62.5/125 micron Multimode fibercable :-

Observation:- 1) Anticlockwise Direction (+ve).

Sr.no.	Angle (degree)	Laser intensity (μW)	In (%)
1	0	39.5	99
2	1	39.6	100
3	2	27.5	69
4	3	24.7	62
5	4	28.5	72
6	5	29.7	75
7	6	4.8	12
8	7	1.1	3
9	8	0.4	1

2) Clockwise Direction (-ve).

Sr.no.	Angle (degree)	Laser intensity (μW)	In (%)
1	0	39.6	100
2	1	32.8	83
3	2	32.2	81
4	3	28.4	72
5	4	25.0	63
6	5	20.6	52
7	6	23.4	59
8	7	2.8	7
9	8	0.6	2

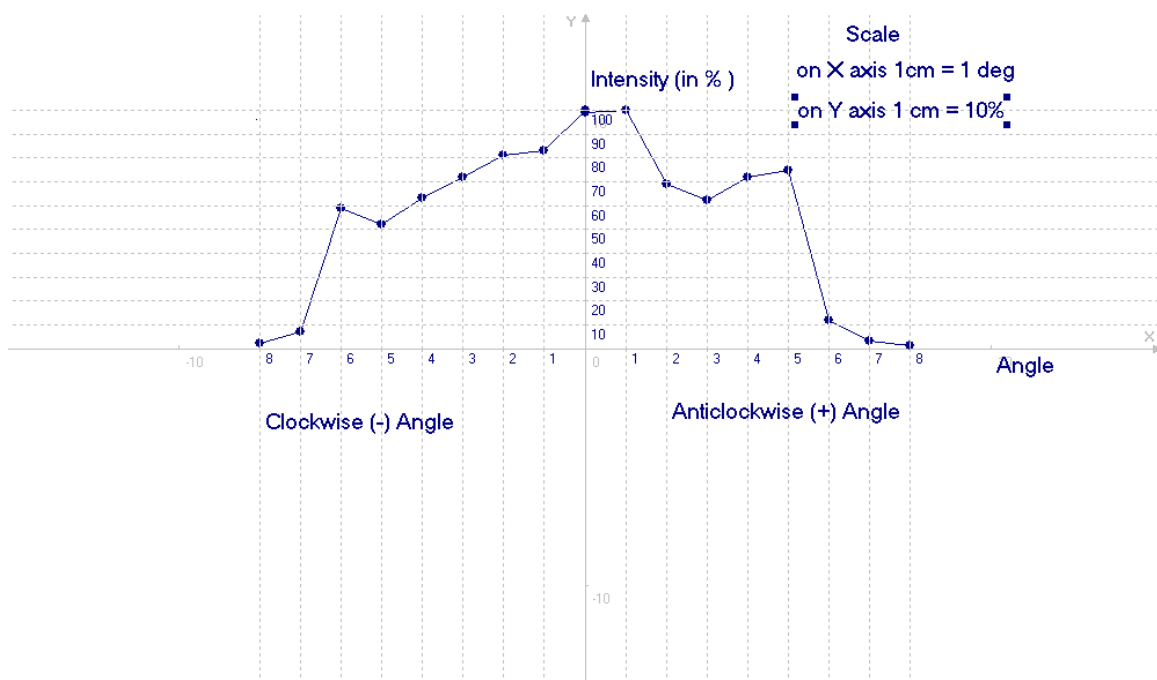


PHOTO DETECTOR UNIT

SPECIFICATIONS

Measuring range	40.00 μ / 400.0 μ / 4.000m / 40.00mW
Wavelength range	400nm to 1100nm
Wavelength	633nm(He-Ne laser) reference wavelength convert by a table of spectral sensitivity characteristic (representing value)
Optical sensor	Si photodiode(Φ 9mm)
Display	Numeric : 3999 full scale, Bargraph : 42 segment
Over display	The character 4 appears in the highest digit
Battery low warning	The character BT appears in the display when the built-in battery is nearly exhausted and battery supply voltage drops
Sampling rate	Approx. 2 times/sec. for numeral display Approx. 20 times/sec. for bar graph.
Operating temperature / Humidity	0°C ~ 40°C, 80%RH(Max) No condensation
Storage temperature / Environmental condition	-10°C ~ 50°C, 80%RH(Max) Altitude 2000m or below, pollution degree II
Power consumption	Approx. 6mW
Safety standards	IEC61326-1
Battery	LR44 x 2
Size / Mass	Main body : H117 x W76 x D18mm/approx. 120g
Sensor probe :	H 84mm x W 16mm x D 10mm

To install photo detector unit in a holder refer to the below figure. Holder is mounted on a adjustable stand. Holder has a U groove. Insert Sensor probe in the U groove with sensing area facing out word.



Fig. : Photo Detector Holder with adjustable stand.

Once sensor probe is placed inside the groove, place appropriate detector cap required by the experiment. Adjust the probe so that active area is exposed to the slit or receptacle on the photo detector cap.

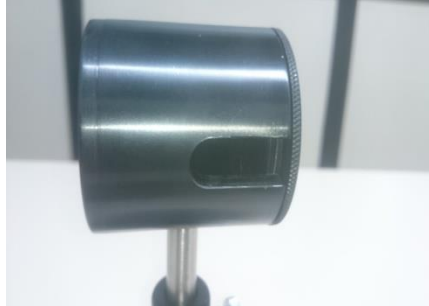


Fig. : Photo Detector Holder After Placing Detector Cap

COMPONENT TOUR

1) OPTICAL BREADBOARD:



Fig. : Optical Breadboard.

2) X-Y- Z TRANSLATION STAGE:



Fig. : XYZ Translation stage.

3) ROTARY STAGE:



Fig. : Rotary stage.

4) ROTARY STAGE WITH RECEPTACLE:

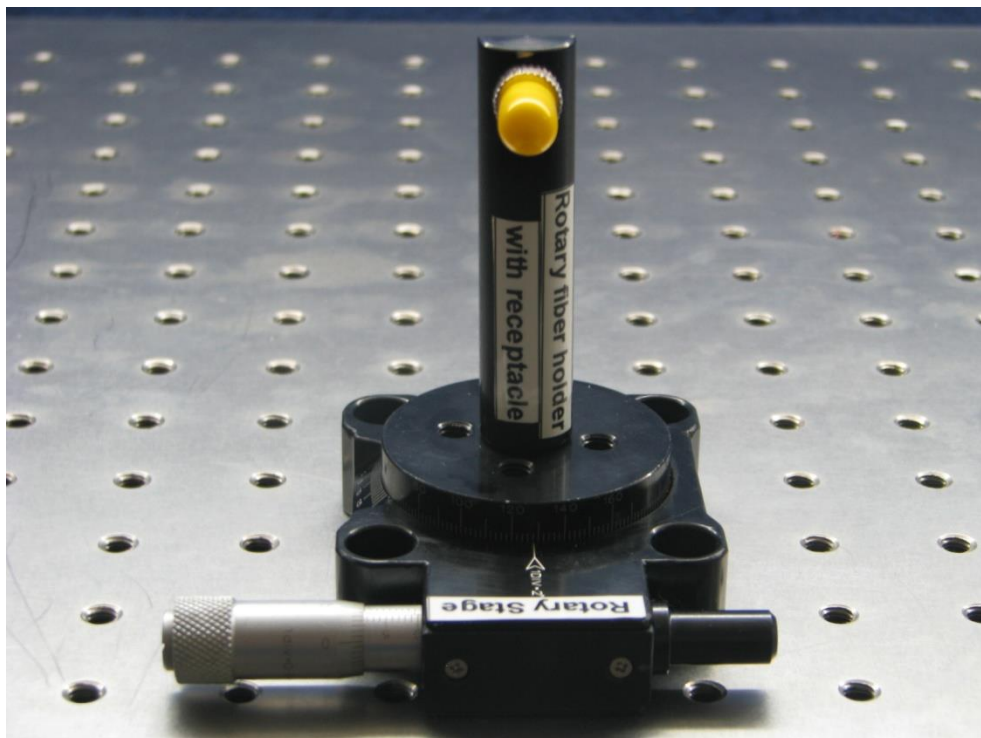


Fig. : Rotary stage with receptacle.

5) GIMBLE MOUNT:

Fig. : Gimble mount.

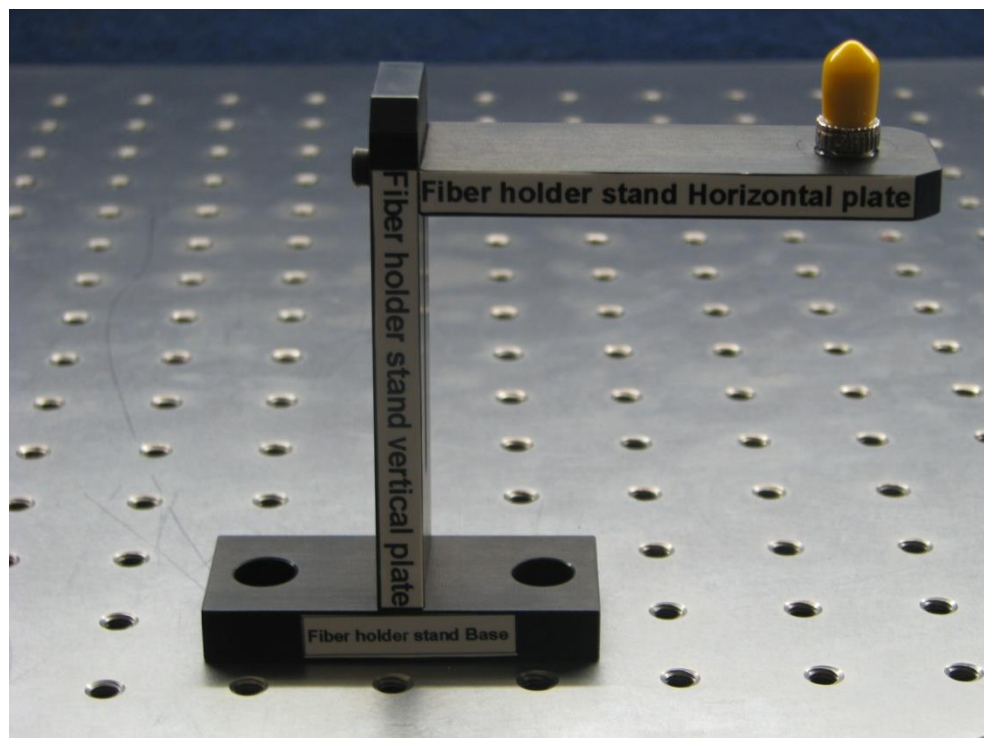
6) FIBER HOLDER STAND:

Fig. : Fiber Holder stand.

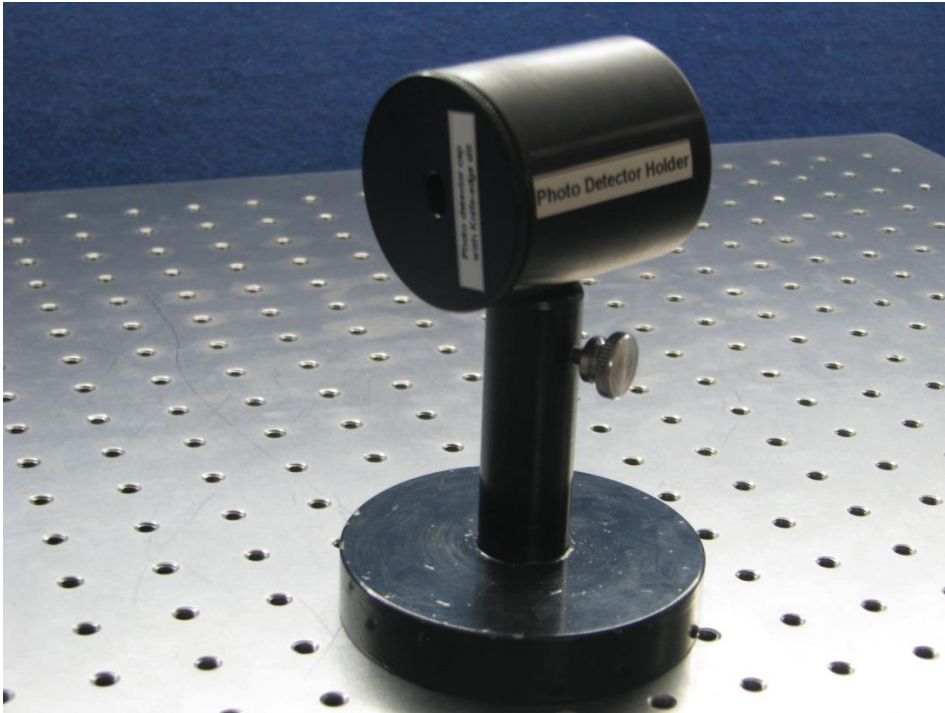
7) PHOTO DETECTOR HOLDER WITH ADJUSTABLE STAND:

Fig. : Photo Detector Holder with adjustable stand.

8) PHOTO DETECTOR CAP WITH RECEPTACLE:

Fig. : Photo Detector cap with receptacle.

9) He-Ne LASER SOURCE WITH POWER SUPPLY':



Fig. : He-Ne Laser source with power supply.

10)OPTICAL POWER METER:



Fig. : Optical Power meter.

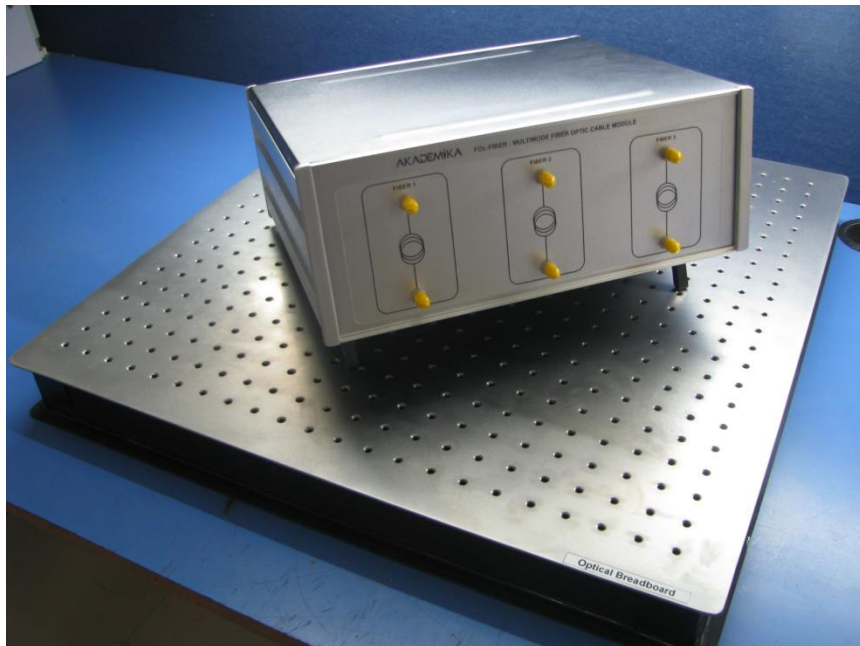
11)FOL-FIBER (100M-500M-1000M MULTIMODE FIBER CABLE SPOOL):

Fig. : FOL-FIBER Spool.

12)LASER TO FIBER COUPLER WITH GROOVE FIBER HOLDER:

Fig. : Laser to Fiber coupler with groove fiber holder.